arXiv:2301.08243v3 [cs.CV] 13 Apr 2023

Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture

Mahmoud Assran[1,2,3*] **Quentin Duval**[1] **Ishan Misra**[1] **Piotr Bojanowski**[1]
Pascal Vincent[1] **Michael Rabbat**[1,3] **Yann LeCun**[1,4] **Nicolas Ballas**[1]

[1]Meta AI (FAIR) [2]McGill University [3] Mila, Quebec AI Institute [4]New York University

Abstract

This paper demonstrates an approach for learning highly semantic image representations without relying on hand-crafted data-augmentations. We introduce the Image-based Joint-Embedding Predictive Architecture (I-JEPA), a non-generative approach for self-supervised learning from images. The idea behind I-JEPA is simple: from a single context block, predict the representations of various target blocks in the same image. A core design choice to guide I-JEPA towards producing semantic representations is the masking strategy; specifically, it is crucial to (a) sample target blocks with sufficiently large scale (semantic), and to (b) use a sufficiently informative (spatially distributed) context block. Empirically, when combined with Vision Transformers, we find I-JEPA to be highly scalable. For instance, we train a ViT-Huge/14 on ImageNet using 16 A100 GPUs in under 72 hours to achieve strong downstream performance across a wide range of tasks, from linear classification to object counting and depth prediction.

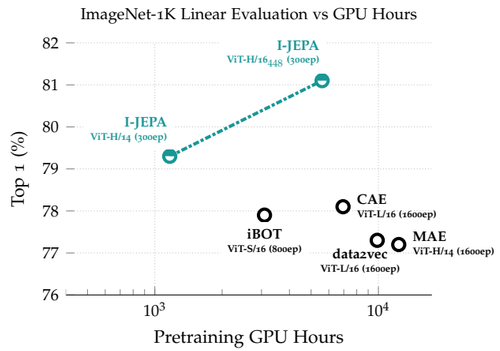


Figure 1. **ImageNet Linear Evaluation**. The I-JEPA method learns semantic image representations without using any view data augmentations during pretraining. By predicting in representation space, I-JEPA produces semantic representations while using less compute than previous methods.

1. Introduction

In computer vision, there are two common families of approaches for self-supervised learning from images: invariance-based methods [1,4,10,17,18,24,35,37,74] and generative methods [8,28,36,57].

Invariance-based pretraining methods optimize an encoder to produce similar embeddings for two or more views of the same image [15, 20], with image views typically constructed using a set of hand-crafted data augmentations, such as random scaling, cropping, and color jittering [20], amongst others [35]. These pretraining methods can produce representations of a high semantic level [4, 18], but they also introduce strong biases that may be detrimental for certain downstream tasks or even for pretraining tasks with different data distributions [2]. Often, it is unclear how to generalize these biases for tasks requiring different levels of abstraction. For example, image classification and instance segmentation do not require the same invariances [11]. Additionally, it is not straightforward to generalize these image-specific augmentations to other modalities such as audio.

Cognitive learning theories have suggested that a driving mechanism behind representation learning in biological systems is the adaptation of an internal model to predict sensory input responses [31, 59]. This idea is at the core of self-supervised generative methods, which remove or corrupt portions of the input and learn to predict the corrupted content [9, 36, 57, 67, 68, 71]. In particular, mask-denoising approaches learn representations by reconstructing randomly masked patches from an input, either at the pixel or token level. Masked pretraining tasks require less prior knowledge than view-invariance approaches and easily generalize beyond the image modality [8]. However, the

*massran@meta.com

arXiv:2301.08243v3 [cs.CV] 13 Apr 2023

通过联合嵌入预测架构进行图像自监督学习

Mahmoud Assran[1],2,[3*] QuentinDuval[1] Ishan Misra[1] Piotr Bojanowski[1]Pascal Vincent[1] Michael Rabbat[1,3] Yann LeCun[1,4] Nicolas Ballas[1]

[1]Meta AI（FAIR） [2]麦吉尔大学 [3] Mila，魁北克人工智能研究所 [4]纽约大学

摘要

本文展示一种无需手工设计数据增强、即可学习高度语义化图像表征的方法。我们提出基于图像的联合嵌入预测架构（*I-JEPA*），一种非生成式图像自监督方法。其思路很简单：从单个上下文块，预测同一图像中多个目标块的表征。引导 *I-JEPA* 学习语义表征的核心设计是掩码策略：必须（*a*）抽取足够大、具有语义的目标块，以及（*b*）使用足够有信息、空间上分布广泛的上下文块。实验发现，结合 *Vision Transformer* 后，*I-JEPA* 具有很好的扩展性。例如，在 *16* 张 *A100* 上不到 *72* 小时即可用 *ImageNet* 训练 *ViT-Huge/14*，在从线性分类到物体计数和深度预测的广泛任务中取得良好性能。

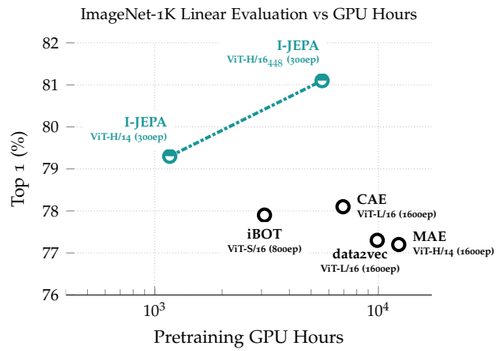


图 1. ImageNet 线性评估。I-JEPA 预训练不使用视图增强，在表征空间预测，以更少计算学习语义图像表征。

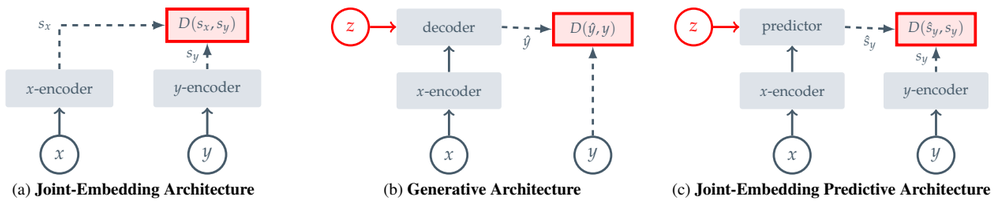
1. 引言

计算机视觉图像自监督学习有两类常见方法：不变性方法 [1,4,10,17,18,24,35,37,74] 和生成式方法 [8,28,36,57]。

不变性预训练优化编码器，为同一图像两个或多个视图生成相似嵌入 [15, 20]，，视图通常通过随机缩放、裁剪、颜色扰动 [20],等手工增强 [35]构造。这些方法可生成高度语义化表征 [4, 18]，，但引入很强偏置，可能损害某些下游任务，或不同数据分布的预训练任务 [2]。我们往往不清楚如何将这些偏置推广到需要不同抽象层次的任务。例如分类与实例分割需要不同不变性 [11]；图像专用增强也不易直接推广至音频等模态。

认知学习理论认为，生物系统中表征学习的驱动机制，是调整内部模型以预测感官输入响应 [31, 59]。这一思想是生成式自监督方法的核心：移除或破坏部分输入，学习预测被破坏内容 [9, 36, 57, 67, 68, 71]。尤其是掩码去噪方法，通过在像素或令牌层重建随机遮蔽图块学习表征。这类任务比视图不变性方法需要更少先验，易推广至图像之外的模态 [8]。但其

*massran@meta.com



(a) **Joint-Embedding Architecture** (b) **Generative Architecture** (c) **Joint-Embedding Predictive Architecture**

Figure 2. Common architectures for self-supervised learning, in which the system learns to capture the relationships between its inputs. The objective is to assign a high energy (large scaler value) to incompatible inputs, and to assign a low energy (low scaler value) to compatible inputs. **(a)** Joint-Embedding Architectures learn to output similar embeddings for compatible inputs x, y and dissimilar embeddings for incompatible inputs. **(b)** Generative Architectures learn to directly reconstruct a signal y from a compatible signal x, using a decoder network that is conditioned on additional (possibly latent) variables z to facilitate reconstruction. **(c)** Joint-Embedding Predictive Architectures learn to predict the embeddings of a signal y from a compatible signal x, using a predictor network that is conditioned on additional (possibly latent) variables z to facilitate prediction.

resulting representations are typically of a lower semantic level and underperform invariance-based pretraining in off-the-shelf evaluations (e.g., linear-probing) and in transfer settings with limited supervision for semantic classification tasks [4]. Consequently, a more involved adaptation mechanism (e.g., end-to-end fine-tuning) is required to reap the full advantage of these methods.

In this work, we explore how to improve the semantic level of self-supervised representations without using extra prior knowledge encoded through image transformations. To that end, we introduce a joint-embedding predictive architecture [48] for images (I-JEPA). An illustration of the method is provided in Figure 3. The idea behind I-JEPA is to predict missing information in an abstract representation space; e.g., given a single context block, predict the representations of various target blocks in the same image, where target representations are computed by a learned target-encoder network.

Compared to generative methods that predict in pixel/token space, I-JEPA makes use of abstract prediction targets for which unnecessary pixel-level details are potentially eliminated, thereby leading the model to learn more semantic features. Another core design choice to guide I-JEPA towards producing semantic representations is the proposed multi-block masking strategy. Specifically, we demonstrate the importance of predicting sufficiently large target blocks in the image, using an informative (spatially distributed) context block.

Through an extensive empirical evaluation, we demonstrate that:

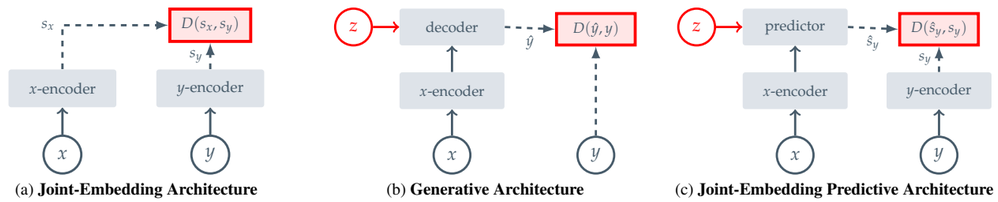
- I-JEPA learns strong off-the-shelf representations without the use of hand-crafted view augmentations (cf. Fig.1). I-JEPA outperforms pixel-reconstruction methods such as MAE [36] on ImageNet-1K linear probing, semi-supervised 1% ImageNet-1K, and semantic transfer tasks.
- I-JEPA is competitive with view-invariant pretraining approaches on semantic tasks and achieves better performance on low-level visions tasks such as object counting and depth prediction (Sections 5 and 6). By using a simpler model with less rigid inductive bias, I-JEPA is applicable to a wider set of tasks.
- I-JEPA is also scalable and efficient (Section 7). Pretraining a ViT-H/14 on ImageNet requires less than 1200 GPU hours, which is over 2.5× faster than a ViT-S/16 pretrained with iBOT [79] and over 10× more efficient than a ViT-H/14 pretrained with MAE. Predicting in representation space significantly reduces the total computation needed for self-supervised pretraining.

2. Background

Self-supervised learning is an approach to representation learning in which a system learns to capture the relationships between its inputs. This objective can be readily described using the framework of Energy-Based Models (EBMs) [49] in which the self-supervised objective is to assign a high energy to incompatible inputs, and to assign a low energy to compatible inputs. Many existing generative and non-generative approaches to self-supervised learning can indeed be cast in this framework; see Figure 2.

Joint-Embedding Architectures. Invariance-based pretraining can be cast in the framework of EBMs using a Joint-Embedding Architecture (JEA), which learns to output similar embeddings for compatible inputs, $\boldsymbol{x}, \boldsymbol{y}$, and dissimilar embeddings for incompatible inputs; see Figure 2a. In the context of image-based pretraining, compatible $\boldsymbol{x}, \boldsymbol{y}$ pairs are typically constructed by randomly applying hand-crafted data augmentations to the same input image [20].

The main challenge with JEAs is representation collapse, wherein the energy landscape is flat (i.e., the encoder produces a constant output regardless of the input). During the past few years, several approaches have been investi-



(a) **Joint-Embedding Architecture** (b) **Generative Architecture** (c) **Joint-Embedding Predictive Architecture**

图 2. 常见自监督架构，学习输入之间的关系。目标是给不兼容输入高能量（较大标量），给兼容输入低能量（较小标量）。（a）联合嵌入架构为兼容输入 x, y 生成相似嵌入，为不兼容输入生成不同嵌入。（b）生成式架构用以额外变量 z（可能为潜变量）为条件的解码器，从兼容信号 x直接重建信号 y。（c）联合嵌入预测架构用以额外变量 z（可能为潜变量）为条件的预测器，从兼容信号 x预测信号 y 的嵌入。

所得表征通常语义层次较低，在直接使用（如线性探测）以及有限监督的语义分类迁移中，落后于不变性预训练 [4]。因此，需要更复杂的适配（如端到端微调），才能发挥这些方法的全部优势。

本文探索如何不借助图像变换所编码的额外先验，提升自监督表征的语义层次。为此引入图像联合嵌入预测架构 [48]（I-JEPA），图 3 给出示意。它在抽象表征空间预测缺失信息，例如从单个上下文块预测同图中的多个目标块表征；目标表征由学得的目标编码器网络计算。

相比在像素／令牌空间预测的生成式方法，I-JEPA 使用可能消除无关像素细节的抽象目标，使模型学习更语义化特征。另一核心设计是多块掩码策略，尤其需要借助有信息、空间分布广泛的上下文，预测足够大的目标块。

通过广泛实证评估，我们证明：

•I-JEPA 无需手工视图增强，即可学习强大、可直接使用的表征（图 1）。在 ImageNet-1K 线性探测、1% 半监督评估及语义迁移中，超过 MAE [36]等像素重建方法。

• I-JEPA 与视图不变性预训练方法相当，

语义任务上的方法，并在物体计数与深度预测等低层视觉任务中更好（第 5、6 节）。模型更简单、归纳偏置较不僵硬，使 I-JEPA 适用于更广泛任务。

• I-JEPA 可扩展且高效（第 7 节）。ImageNet 上 ViT-H/14 预训练少于 1200 GPU 小时，比 iBOT [79] 的 ViT-S/16 快超过 2.5×，比 MAE 预训练 ViT-H/14 高效超过 10×。在表征空间预测显著减少自监督预训练总计算量。

2. 背景

自监督学习是一种表征学习，系统学习捕获输入间关系。这可通过能量模型（EBM）[49] 描述：为不兼容输入赋高能量，为兼容输入赋低能量。许多生成式和非生成式自监督方法都可纳入此框架，见图 2。

联合嵌入架构。不变性预训练可通过 JEA 纳入能量模型框架，对兼容输入 $\boldsymbol{x}, \boldsymbol{y}$输出相似嵌入，不兼容输入输出不同嵌入（图 2a）。图像预训练中，通常对同一图像随机使用手工增强 [20]，构造兼容的 $\boldsymbol{x}, \boldsymbol{y}$对。

联合嵌入架构的主要挑战是表征坍塌，能量景观变平，即编码器无论输入都输出常量。过去几年已探

gated to prevent representation collapse, such as contrastive losses that explicitly push apart embeddings of negative examples [15,24,37], non-contrastive losses that minimize the informational redundancy across embeddings [10, 74], and clustering-based approaches that maximize the entropy of the average embedding [4, 5, 18]. There are also heuristic approaches that leverage an asymmetric architectural design between the x-encoder and y-encoder to avoid collapse [8, 24, 35].

Generative Architectures. Reconstruction-based methods for self-supervised learning can also be cast in the framework of EBMs using Generative Architectures; see Figure 2b. Generative Architectures learn to directly reconstruct a signal $\boldsymbol{y}$ from a compatible signal $\boldsymbol{x}$, using a decoder network that is conditioned on an additional (possibly latent) variable $\boldsymbol{z}$ to facilitate reconstruction. In the context of image-based pretraining, one common approach in computer vision is to produce compatible $\boldsymbol{x}, \boldsymbol{y}$ pairs using masking [9, 38] where $\boldsymbol{x}$ is a copy of the image $\boldsymbol{y}$, but with some of the patches masked. The conditioning variable $\boldsymbol{z}$ then corresponds to a set of (possibly learnable) mask and position tokens, that specifies to the decoder which image patches to reconstruct. Representation collapse is not a concern with these architectures as long as the informational capacity of $\boldsymbol{z}$ is low compared to the signal $\boldsymbol{y}$.

Joint-Embedding Predictive Architectures. As shown in Figure 2c, Joint-Embedding Predictive Architectures [48] are conceptually similar to Generative Architectures; however, a key difference is that the loss function is applied in embedding space, not input space. JEPAs learn to predict the embeddings of a signal $\boldsymbol{y}$ from a compatible signal $\boldsymbol{x}$, using a predictor network that is conditioned on an additional (possibly latent) variable $\boldsymbol{z}$ to facilitate prediction. Our proposed I-JEPA provides an instantiation of this architecture in the context of images using masking; see Figure 3.

In contrast to Joint-Embedding Architectures, JEPAs do not seek representations invariant to a set of hand-crafted data augmentations, but instead seek representations that are predictive of each other when conditioned on additional information $\boldsymbol{z}$. However, as with Joint-Embedding Architectures, representation collapse is also a concern with JEPAs; we leverage an asymmetric architecture between the $\boldsymbol{x}$- and $\boldsymbol{y}$-encoders to avoid representation collapse.

3. Method

We now describe the proposed Image-based Joint-Embedding Predictive Architecture (I-JEPA), illustrated in Figure 3. The overall objective is as follows: given a context block, predict the representations of various target blocks

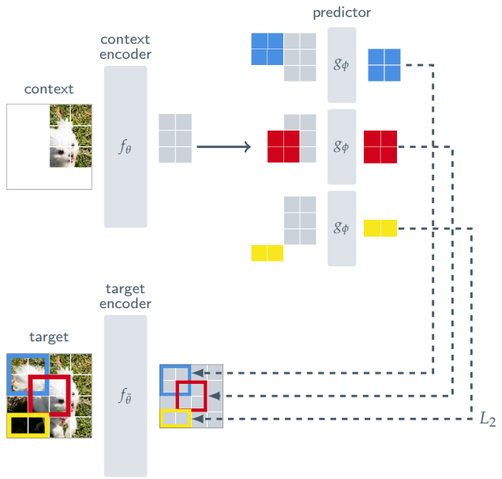


Figure 3. **I-JEPA.** The Image-based Joint-Embedding Predictive Architecture uses a single context block to predict the representations of various target blocks originating from the same image. The context encoder is a Vision Transformer (ViT), which only processes the visible context patches. The predictor is a narrow ViT that takes the context encoder output and, conditioned on positional tokens (shown in color), predicts the representations of a target block at a specific location. The target representations correspond to the outputs of the target-encoder, the weights of which are updated at each iteration via an exponential moving average of the context encoder weights.

in the same image. We use a Vision Transformer [29, 63] (ViT) architecture for the context-encoder, target-encoder, and predictor. A ViT is composed of a stack of transformer layers, each consisting of a self-attention [66] operation followed by a fully-connected MLP. Our encoder/predictor architecture is reminiscent of the generative masked autoencoders (MAE) [36] method. However, one key difference is that the I-JEPA method is non-generative and the predictions are made in representation space.

Targets. We first describe how we produce the targets in the I-JEPA framework: in I-JEPA, the targets correspond to the representations of image blocks. Given an input image $\boldsymbol{y}$, we convert it into a sequence of N non-overlapping patches, and feed this through the target-encoder $f_{\bar{\theta}}$ to obtain a corresponding patch-level representation $\boldsymbol{s}_y = \{\boldsymbol{s}_{y_1}, \ldots, \boldsymbol{s}_{y_N}\}$ where $\boldsymbol{s}_{y_k}$ is the representation associated with the kth patch. To obtain the targets for our loss, we randomly sample M (possibly overlapping) blocks from the target representations $\boldsymbol{s}_y$. We denote by B_i the mask corresponding of the ith block and by $\boldsymbol{s}_y(i) = \{\boldsymbol{s}_{y_j}\}_{j \in B_i}$ its

索过多种防坍塌方法，如显式推开负样本嵌入的对比损失 [15,24,37], 、降低嵌入信息冗余的非对比损失 [10, 74], ，以及最大化平均嵌入熵的聚类方法 [4, 5, 18]。还有启发式方法，通过 x 编码器与 y编码器间非对称架构避免坍塌 [8, 24, 35]。

生成式架构。重建式自监督方法也可通过生成式架构纳入能量模型框架（图 2b）。它们通过以额外变量 $\boldsymbol{z}$ （可能是潜变量）为条件的解码器，从兼容信号 $\boldsymbol{x}$直接重建信号 $\boldsymbol{y}$ 。视觉预训练常通过遮蔽 [9, 38] 构造兼容的 $\boldsymbol{x}, \boldsymbol{y}$ 对，其中 $\boldsymbol{x}$ 是图像 $\boldsymbol{y}$的副本，部分图块被遮蔽。条件变量 $\boldsymbol{z}$是掩码和位置令牌（可能可学习）的集合，告诉解码器要重建哪些图块。只要 $\boldsymbol{z}$ 的信息容量相对信号 $\boldsymbol{y}$较低，这类架构就无需担心表征坍塌。

联合嵌入预测架构。如图 2c，JEPA [48]在概念上类似生成式架构，但关键区别是损失作用于嵌入空间，而非输入空间。以额外变量 $\boldsymbol{z}$ （可能为潜变量）为条件的预测器，从兼容信号 $\boldsymbol{x}$预测信号 $\boldsymbol{y}$ 的嵌入。I-JEPA 用图像遮蔽实例化此架构，见图 3。

与联合嵌入架构不同，JEPA 不追求对一组手工增强保持不变的表征，而追求在额外信息条件下能相互预测的表征 $\boldsymbol{z}$。但它同样存在坍塌风险，我们通过$\boldsymbol{x}$和 $\boldsymbol{y}$编码器的非对称架构避免坍塌。

3. 方法

现在介绍图 3 的基于图像联合嵌入预测架构（I-JEPA）。整体目标是：给定上下文块，预测多个目标块的表征，

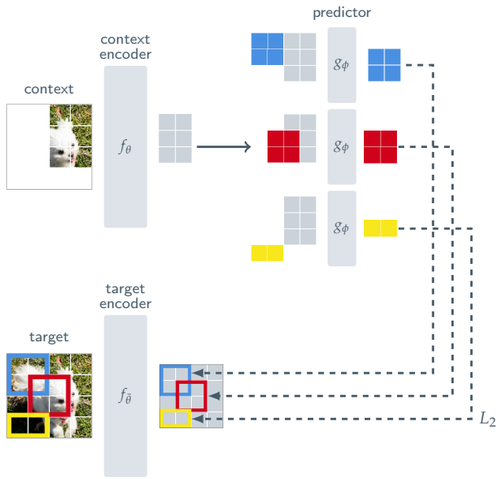


图 3. I-JEPA 从单个上下文块预测同一图像中多个目标块的表征。上下文编码器是只处理可见图块的 ViT。预测器是较窄 ViT，接收上下文输出，并以位置令牌（彩色）为条件预测指定位置的目标块表征。目标来自目标编码器输出，其权重每次迭代通过上下文编码器权重的指数移动平均更新。

在同一图像中。上下文编码器、目标编码器与预测器均采用 Vision Transformer（ViT） [29, 63]。ViT 由 Transformer 层堆叠组成，每层自注意力 [66]后接全连接 MLP。编码器／预测器类似生成式 MAE [36]架构，但关键区别是 I-JEPA 非生成式，在表征空间预测。

目标。先描述 I-JEPA 如何生成目标：目标是图像块的表征。将输入图像 $\boldsymbol{y}$转换为 N 个不重叠图块，送入目标编码器 $f_{\bar{\theta}}$ ，获得图块级表征 $\boldsymbol{s}_y = \{\boldsymbol{s}_{y_1}, \ldots, \boldsymbol{s}_{y_N}\}$，其中 $\boldsymbol{s}_{y_k}$ 是第 k^{th} 个图块的表征。为构造损失目标，从目标表征 $\boldsymbol{s}_y$随机抽取 M 个可能重叠的块。 B_i 表示第 i^{th} 个块的掩码， $\boldsymbol{s}_y(i) = \{\boldsymbol{s}_{y_j}\}_{j \in B_i}$ 表示它的



Figure 4. Examples of our context and target-masking strategy. Given an image, we randomly sample 4 target blocks with scale in the range $(0.15, 0.2)$ and aspect ratio in the range $(0.75, 1.5)$. Next, we randomly sample a context block with scale in the range $(0.85, 1.0)$ and remove any overlapping target blocks. Under this strategy, the target-blocks are relatively semantic, and the context-block is informative, yet sparse (efficient to process).

patch-level representation. Typically, we set M equal to 4, and sample the blocks with a random aspect ratio in the range $(0.75, 1.5)$ and random scale in the range $(0.15, 0.2)$. Note that the target blocks are obtained by masking the *output* of the target-encoder, not the input. This distinction is crucial to ensure target representations of a high semantic level; see, e.g., [8].

Context. Recall, the goal behind I-JEPA is to predict the target block representations from a single context block. To obtain the context in I-JEPA, we first sample a single block $\boldsymbol{x}$ from the image with a random scale in the range $(0.85, 1.0)$ and unit aspect ratio. We denote by B_x the mask associated with the context block $\boldsymbol{x}$. Since the target blocks are sampled independently from the context block, there may be significant overlap. To ensure a non-trivial prediction task, we remove any overlapping regions from the context block. Figure 4 shows examples of various context and target blocks in practice. Next, the masked context block, $\boldsymbol{x}$, is fed through the context encoder f_θ to obtain a corresponding patch-level representation $\boldsymbol{s}_x = \{\boldsymbol{s}_{x_j}\}_{j \in B_x}$.

Prediction. Given the output of the context encoder, $\boldsymbol{s}_x$, we wish to predict the M target block representations $\boldsymbol{s}_y(1), \ldots, \boldsymbol{s}_y(M)$. To that end, for a given target block $\boldsymbol{s}_y(i)$ corresponding to a target mask B_i, the predictor $g_\phi(\cdot, \cdot)$ takes as input the output of the context encoder $\boldsymbol{s}_x$ and a mask token for each patch we wish to predict, $\{\boldsymbol{m}_j\}_{j \in B_i}$, and outputs a patch-level prediction $\hat{\boldsymbol{s}}_y(i) = \{\hat{\boldsymbol{s}}_{y_j}\}_{j \in B_i} = g_\phi(\boldsymbol{s}_x, \{\boldsymbol{m}_j\}_{j \in B_i})$. The mask tokens are parameterized by a shared learnable vector with an added positional embedding. Since we wish to make predictions for M target blocks, we apply our predictor M times, each time conditioning on the mask tokens corresponding to the target-block locations we wish to predict, and obtain predictions $\hat{\boldsymbol{s}}_y(1), \ldots, \hat{\boldsymbol{s}}_y(M)$.

Loss. The loss is simply the average L_2 distance between the predicted patch-level representations $\hat{\boldsymbol{s}}_y(i)$ and the target patch-level representation $\boldsymbol{s}_y(i)$; i.e.,

$$\frac{1}{M} \sum_{i=1}^{M} D\left(\hat{\boldsymbol{s}}_y(i), \boldsymbol{s}_y(i)\right) = \frac{1}{M} \sum_{i=1}^{M} \sum_{j \in B_i} \|\hat{\boldsymbol{s}}_{y_j} - \boldsymbol{s}_{y_j}\|_2^2.$$

The parameters of the predictor, ϕ, and context encoder, θ, are learned through gradient-based optimization, while the parameters of the target encoder $\bar{\theta}$ are updated via an exponential moving average of the context-encoder parameters. The use of an exponential moving average target-encoder has proven essential for training JEAs with Vision Transformers [18, 25, 79], we find the same to be true for I-JEPA.

4. Related Work

A long line of work has explored visual representation learning by predicting the values of missing or corrupted sensory inputs. Denoising autoencoders use random noise as input corruption [67]. Context encoders regress an entire image region based on its surrounding [57]. Other works cast image colorization as a denoising task [46, 47, 77].

The idea of image denoising has recently been revisited in the context of masked image modelling [9, 36, 71], where a Vision Transformer [29] is used to reconstruct missing input patches. The work on Masked Autoencoders (MAE) [36] proposed an efficient architecture that only requires the encoder to process visible image patches. By reconstructing missing patches in pixels space, MAE achieves strong performance when fine-tuned end-to-end on large labeled datasets and exhibits good scaling properties. BEiT [9] predicts the value of missing patches in a tokenized space; specifically, tokenizing image patches using a frozen discreteVAE, which is trained on a dataset containing 250 million images [58]. Yet, pixel-level pre-training has been shown to outperform BEiT for fine-tuning [36]. Another work, SimMIM [71], explores reconstruction targets based on the classic Histogram of Gradients [27] feature space, and demonstrates some advantage over pixel space reconstruction. Different from those works, our representation space is learned during training through a Joint-Embedding Predictive Architecture. Our goal is to learn semantic representations that do not require extensive fine-tuning on downstream tasks.

Closest to our work is data2vec [8] and Context Autoencoders [25]. The data2vec method learns to predict the rep-



图 4. 上下文与目标掩码策略示例。先随机抽取 4 个目标块，尺度（0.15, 0.2），纵横比（0.75, 1.5），再抽取尺度（0.85, 1.0）的上下文块，并移除与目标重叠的部分。目标块较有语义，上下文信息充分且稀疏，因此处理高效。

图块级表征。通常 M 设为 4，目标块纵横比随机取（0.75, 1.5），尺度取（0.15, 0.2）。目标块通过遮蔽目标编码器输出而非输入获得，这一差异对确保高语义目标至关重要（如 [8]）。

上下文。I-JEPA 从单个上下文块预测目标块表征。先从图像抽取上下文块 $\boldsymbol{x}$，尺度随机取（0.85, 1.0），纵横比为 1；B_x 表示块 $\boldsymbol{x}$的掩码。目标与上下文独立抽样，可能大幅重叠。为避免平凡预测，移除上下文中所有重叠区域。图 4 给出实际例子。接着将掩码上下文块$\boldsymbol{x}$送入上下文编码器 f_θ，取得图块级表征 $\boldsymbol{s}_x = \{\boldsymbol{s}_{x_j}\}_{j \in B_x}$。

预测。给定上下文编码器输出 $\boldsymbol{s}_x$，希望预测 M 个目标块表征 $\boldsymbol{s}_y(1) \ldots, \boldsymbol{s}_y(M)$。对目标掩码 B_i对应的块$\boldsymbol{s}_y(i)$，预测器$g_\phi(\cdot, \cdot)$ 接收上下文编码器输出$\boldsymbol{s}_x$ 及每个待预测图块的掩码令牌 $\{\boldsymbol{m}_j\}_{j \in B_i}$，输出图块级预测$\hat{}\ \boldsymbol{s}_y(i) = \{\hat{}\boldsymbol{s}_{y_j}\}_{j \in B_i} = g_\phi(\boldsymbol{s}_x, \{\boldsymbol{m}_j\}_{j \in B_i})$。掩码令牌

由共享可学习向量加位置嵌入参数化。为预测 M 个目标块，运行预测器 M 次，每次以所需目标位置的掩码令牌为条件，得到预测$\hat{}\ \boldsymbol{s}_y(1) \ldots, \hat{}\boldsymbol{s}_y(M)$。

损失。损失是预测图块表征$\hat{}\ \boldsymbol{s}_y(i)$与目标图块表征 $\boldsymbol{s}_y(i)$之间的平均 L_2 距离，即：

$$\frac{1}{M} \sum_{i=1}^{M} D\left(\hat{\boldsymbol{s}}_y(i), \boldsymbol{s}_y(i)\right) = \frac{1}{M} \sum_{i=1}^{M} \sum_{j \in B_i} \|\hat{\boldsymbol{s}}_{y_j} - \boldsymbol{s}_{y_j}\|_2^2.$$

预测器参数 ϕ和上下文编码器参数 θ通过梯度优化学习；目标编码器 θ 的¯参数通过上下文编码器参数的指数移动平均更新。对 ViT 联合嵌入架构，EMA 目标编码器已被证明至关重要 [18, 25, 79]，；I-JEPA 也如此。

4. 相关工作

许多工作通过预测缺失或受损感官输入的值来学习视觉表征。去噪自编码器以随机噪声破坏输入 [67]，上下文编码器从周边回归整块图像区域 [57]，另有工作将图像着色视为去噪任务 [46, 47, 77]。

图像去噪最近在掩码图像建模 [9, 36, 71],中重新受到关注，用 Vision Transformer [29] 重建缺失输入图块。掩码自编码器（MAE） [36] 提出高效架构，编码器只需处理可见图块。MAE 在像素空间重建缺失图块，能在大型标注数据集上端到端微调获得良好性能，并有良好扩展性。BEiT [9] 在令牌化空间预测缺失图块，使用冻结的离散 VAE 将图块令牌化，该 VAE 在 2.5 亿张图像上训练 [58]。但像素级预训练的微调表现已被证明优于 BEiT [36]。SimMIM [71], 探索经典梯度直方图 [27] 特征空间的重建目标，相对像素重建有一定优势。与这些工作不同，我们的表征空间通过联合嵌入预测架构在训练中学习。目标是获得无需在下游任务大量微调的语义表征。

最接近本工作的 data2vec [8] 和 Context Autoencoder [25]中，data2vec 学习预测

Method	Arch.	Epochs	Top-1
Methods without view data augmentations			
data2vec [8]	ViT-L/16	1600	77.3
MAE [36]	ViT-B/16	1600	68.0
	ViT-L/16	1600	76.0
	ViT-H/14	1600	77.2
CAE [22]	ViT-B/16	1600	70.4
	ViT-L/16	1600	78.1
I-JEPA	ViT-B/16	600	72.9
	ViT-L/16	600	77.5
	ViT-H/14	300	79.3
	ViT-H/16_{448}	300	**81.1**
Methods using extra view data augmentations			
SimCLR v2 [21]	RN152 (2×)	800	79.1
DINO [18]	ViT-B/8	300	80.1
iBOT [79]	ViT-L/16	250	**81.0**

Table 1. **ImageNet**. Linear-evaluation on ImageNet-1k (the ViT-H/16_{448} is pretrained at at a resolution of 448×448). I-JEPA improves linear probing performance compared to other methods that do not rely on hand-crafted view data-augmentations during pretraining. Moreover, I-JEPA demonstrates good scalability — the larger I-JEPA model matches the performance of view-invariance approaches without requiring view data-augmentations.

resentation of missing patches computed through an online target encoder; by avoiding handcrafted augmentations, the method can be applied to diverse modalities with promising results in vision, text and speech. Context Autoencoders use an encoder/decoder architecture optimized via the sum of a reconstruction loss and an alignment constraint, which enforces predictability of missing patches in representation space. Compared to these methods, I-JEPA exhibits significant improvements in computational efficiency and learns more semantic off-the-shelf representations. Concurrent to our work, data2vec-v2 [7] explores efficient architectures for learning with various modalities.

We also compare I-JEPA with various methods based on joint-embedding architectures; e.g., DINO [18], MSN [4] and iBOT [79]. Theses methods rely on hand-crafted data augmentations during pretraining to learn semantic image representations. The work on MSN [4], uses masking as an additional data-augmentation during pretraining, while iBOT combines a data2vec-style patch-level reconstruction loss with the DINO view-invariance loss. Common to these approaches is the need to process multiple user-generated views of each input image, thereby hindering scalability. By contrast, I-JEPA only requires processing a single view of each image. We find that a ViT-Huge/14 trained with I-JEPA requires less computational effort than a ViT-Small/16 trained with iBOT.

Method	Arch.	Epochs	Top-1
Methods without view data augmentations			
data2vec [8]	ViT-L/16	1600	73.3
MAE [36]	ViT-L/16	1600	67.1
	ViT-H/14	1600	71.5
I-JEPA	ViT-L/16	600	69.4
	ViT-H/14	300	73.3
	ViT-H/16_{448}	300	**77.3**
Methods using extra view data augmentations			
iBOT [79]	ViT-B/16	400	69.7
DINO [18]	ViT-B/8	300	70.0
SimCLR v2 [35]	RN151 (2×)	800	70.2
BYOL [35]	RN200 (2×)	800	71.2
MSN [4]	ViT-B/4	300	**75.7**

Table 2. **ImageNet-1%**. Semi-supervised evaluation on ImageNet-1K using only 1% of the available labels. Models are adapted via fine-tuning or linear-probing, depending on whichever works best for each respective method. ViT-H/16_{448} is pretrained at at a resolution of 448×448. I-JEPA pretraining outperforms MAE which also does not rely on hand-crafted data-augmentations during pretraining. Moreover, I-JEPA benefits from scale. A ViT-H/16 trained at resolution 448 surpasses previous methods including methods that leverage extra hand-crafted data-augmentations.

5. Image Classification

To demonstrate that I-JEPA learns high-level representations without relying on hand-crafted data-augmentations, we report results on various image classification tasks using the linear probing and partial fine-tuning protocols. In this section, we consider self-supervised models that have been pretrained on the ImageNet-1K dataset [60]. Pretraining and evaluation implementation details are described in the Appendix A. All I-JEPA models are trained at resolution 224×224 pixels, unless stated otherwise.

ImageNet-1K. Table 1 shows performance on the common ImageNet-1K linear-evaluation benchmark. After self-supervised pretraining, the model weights are frozen and a linear classifier is trained on top using the full ImageNet-1K training set. Compared to popular methods such as Masked Autoencoders (MAE) [36], Context Autoencoders (CAE) [22], and data2vec [8], which also do not rely on extensive hand-crafted data-augmentations during pretraining, we see that I-JEPA significantly improves linear probing performance, while using less computational effort (see section 7). By leveraging the improved efficiency of I-JEPA, we can train larger models that outperform the best CAE model while using a fraction of the compute. I-JEPA also benefits from scale; in particular, a ViT-H/16 trained at resolution 448×448 pixels matches the performance of view-

方法	架构	轮数	Top-1
不使用视图数据增强的方法			
data2vec [8]	ViT-L/16	1600	77.3
MAE [36]	ViT-B/16	1600	68.0
	ViT-L/16	1600	76.0
	ViT-H/14	1600	77.2
CAE [22]	ViT-B/16	1600	70.4
	ViT-L/16	1600	78.1
I-JEPA	ViT-B/16	600	72.9
	ViT-L/16	600	77.5
	ViT-H/14	300	79.3
	ViT-H/16_{448}	300	**81.1**
额外使用视图数据增强的方法			
SimCLR v2 [21]	RN152 (2×)	800	79.1
DINO [18]	ViT-B/8	300	80.1
iBOT [79]	ViT-L/16	250	**81.0**

表 1. ImageNet-1k 线性评估（ViT-H/16_{448} 以 448×448分辨率预训练）。相较预训练不依赖手工视图增强的方法，I-JEPA 提高线性探测性能。扩展性良好：更大模型无需视图增强便匹配视图不变性方法。

缺失图块的表征由在线目标编码器计算。由于避免手工增强，该方法可用于不同模态，在视觉、文本与语音中都有良好结果。Context Autoencoder 使用编码器／解码器，以重建损失和对齐约束之和优化，约束缺失图块在表征空间可预测。相较之下，I-JEPA 计算效率显著提高，学到更语义化、可直接使用的表征。与本工作同时，data2vec-v2 [7]也探索了多模态高效学习架构。

还与 DINO [18], 、MSN [4]和 iBOT [79]等联合嵌入方法比较，它们通过手工增强学习语义表征。MSN [4], 将遮蔽作为额外增强，iBOT 则结合 data2vec 风格的图块级重建损失和 DINO 视图不变性损失。它们都需处理每张输入图的多个人工生成视图，限制扩展性。I-JEPA 只需一个视图。用 I-JEPA 训练 ViT-Huge/14，计算量少于用 iBOT 训练 ViT-Small/16。

方法	架构	轮数	Top-1
不使用视图数据增强的方法			
data2vec [8]	ViT-L/16	1600	73.3
MAE [36]	ViT-L/16	1600	67.1
	ViT-H/14	1600	71.5
I-JEPA	ViT-L/16	600	69.4
	ViT-H/14	300	73.3
	ViT-H/16_{448}	300	**77.3**
额外使用视图数据增强的方法			
iBOT [79]	ViT-B/16	400	69.7
DINO [18]	ViT-B/8	300	70.0
SimCLR v2 [35]	RN151 (2×)	800	70.2
BYOL [35]	RN200 (2×)	800	71.2
MSN [4]	ViT-B/4	300	**75.7**

表 2. ImageNet-1%。仅用 1% 标签进行半监督评估，各方法选择更好的微调或线性探测。ViT-H/16_{448} 以 448×448分辨率预训练。I-JEPA 优于同样不依赖手工增强的 MAE；扩大规模进一步获益，448 分辨率的 ViT-H/16 超过包括使用额外手工增强的方法。

5. 图像分类

为证明 I-JEPA 无需手工增强也能学到高层表征，我们用线性探测与部分微调，报告各图像分类任务结果。这里采用在 ImageNet-1K [60]预训练的自监督模型。预训练与评估实现见附录 A。除特别说明外，全部 I-JEPA 以224×224 像素分辨率训练。

ImageNet-1K。表 1 展示常用线性评估基准性能。自监督预训练后冻结模型，用完整 ImageNet-1K 训练集训练线性分类器。相比同样不依赖大量手工增强的 MAE [36], 、CAE [22], 和 data2vec [8], , I-JEPA 以更少计算量显著提高线性探测性能（见第 7 节）。利用其效率，可以用少量计算训练更大模型，超过最佳 CAE 模型。I-JEPA 也受益于规模：尤其是以 448×448 像素分辨率训练的 ViT-H/16，可以匹配基于视图

Method	Arch.	CIFAR100	Places205	iNat18
Methods without view data augmentations				
data2vec [8]	ViT-L/16	81.6	54.6	28.1
MAE [36]	ViT-H/14	77.3	55.0	32.9
I-JEPA	ViT-H/14	**87.5**	**58.4**	**47.6**
Methods using extra view data augmentations				
DINO [18]	ViT-B/8	84.9	57.9	55.9
iBOT [79]	ViT-L/16	**88.3**	**60.4**	**57.3**

Table 3. **Linear-probe transfer for image classification**. Linear-evaluation on downstream image classification tasks. I-JEPA significantly outperforms previous methods that also do not use augmentations (MAE and data2vec), and decreases the gap with the best view-invariance-based methods that leverage hand-crafted data augmentations during pretraining.

Method	Arch.	Clevr/Count	Clevr/Dist
Methods without view data augmentations			
data2vec [8]	ViT-L/16	85.3	71.3
MAE [36]	ViT-H/14	**90.5**	**72.4**
I-JEPA	ViT-H/14	86.7	**72.4**
Methods using extra data augmentations			
DINO [18]	ViT-B/8	86.6	53.4
iBOT [79]	ViT-L/16	85.7	62.8

Table 4. **Linear-probe transfer for low-level tasks**. Linear-evaluation on downstream low-level tasks consisting of object counting (Clevr/Count) and depth prediction (Clevr/Dist). The I-JEPA method effectively captures low-level image features during pretraining and outperforms view-invariance based methods on tasks such object counting and depth prediction.

invariant approaches such as iBOT [79], despite avoiding the use of hand-crafted data-augmentations.

Low-Shot ImageNet-1K. Table 2 shows performance on the 1% ImageNet benchmark. Here the idea is to adapt the pretrained models for ImageNet classification using only 1% of the available ImageNet labels, corresponding to roughly 12 or 13 images per class. Models are adapted via fine-tuning or linear-probing, depending on whichever works best for each respective method. I-JEPA outperforms MAE while requiring less pretraining epochs when using a similar encoder architecture. I-JEPA, using a ViT-H/14 architecture, matches the performance of a ViT-L/16 pretrained with data2vec [8], while using significantly less computational effort (see Section 7). By increasing the image input resolution, I-JEPA outperforms previous methods including joint-embedding methods that do leverage extra hand-crafted data-augmentations during pretraining, such as MSN [4], DINO [17], and iBOT [79].

Transfer learning. Table 3 shows performance on various downstream image classification tasks using a linear probe. I-JEPA significantly outperforms previous methods that do not use augmentations (MAE and data2vec), and decreases the gap with the best view-invariance-based methods, which leverage hand-crafted data augmentations during pretraining, even surpassing the popular DINO [18] on CIFAR100 and Place205 with a linear probe.

6. Local Prediction Tasks

As demonstrated in Section 5, I-JEPA learns semantic image representations that significantly improve the downstream image classification performance of previous methods, such as MAE and data2vec. Additionally, I-JEPA benefits from scale and can close the gap, and even surpass, view-invariance based methods that leverage extra hand-crafted data augmentations. In this section, we find that I-JEPA also learns local image features and surpasses view-invariance based methods on low-level and dense prediction tasks, such as object counting and depth prediction.

Table 4 shows performance on various low-level tasks using a linear probe. After pretraining, the encoder weights are frozen and a linear model is trained on top to perform object-counting and depth prediction on the Clevr dataset [43]. Compared to view-invariance methods such as DINO and iBOT, the I-JEPA method effectively captures low-level image features during pretraining and outperforms them in object counting (Clevr/Count) and (by a large margin) depth prediction (Clevr/Dist).

7. Scalability

Model Efficiency. I-JEPA is highly scalable compared to previous approaches. Figure 5 shows semi-supervised evaluation on 1% ImageNet-1K as a function of GPU hours. I-JEPA requires less compute than previous methods and achieves strong performance without relying on hand-crafted data-augmentations. Compared to reconstruction-based methods, such as MAE, which directly use pixels as targets, I-JEPA introduces extra overhead by computing targets in representation space (about 7% slower time per iteration). However, since I-JEPA converges in roughly $5\times$ fewer iterations, we still see significant compute savings in practice. Compared to view-invariance based methods, such as iBOT, which rely on hand-crafted data augmentations to create and process multiple views of each image, I-JEPA also runs significantly faster. In particular, a huge I-JEPA model (ViT-H/14) requires less compute than a small iBOT model (ViT-S/16).

Scaling data size. We also find I-JEPA to benefit from pretraining with larger datasets. Table 5 shows transfer

方法	架构	CIFAR100	Places205	iNat18
不使用视图数据增强的方法				
data2vec [8]	ViT-L/16	81.6	54.6	28.1
MAE [36]	ViT-H/14	77.3	55.0	32.9
I-JEPA	ViT-H/14	**87.5**	**58.4**	**47.6**
额外使用视图数据增强的方法				
DINO [18]	ViT-B/8	84.9	57.9	55.9
iBOT [79]	ViT-L/16	**88.3**	**60.4**	**57.3**

表 3. 下游图像分类的线性探测迁移。I-JEPA 显著超过同样不用增强的 MAE、data2vec，并缩小与预训练使用手工增强的最佳视图不变性方法的差距。

方法	架构	Clevr/Count	Clevr/Dist
不使用视图数据增强的方法			
data2vec [8]	ViT-L/16	85.3	71.3
MAE [36]	ViT-H/14	**90.5**	**72.4**
I-JEPA	ViT-H/14	86.7	**72.4**
使用额外数据增强的方法			
DINO [18]	ViT-B/8	86.6	53.4
iBOT [79]	ViT-L/16	85.7	62.8

表 4. 低层任务线性探测迁移，包括物体计数（Clevr/Count）和深度预测（Clevr/Dist）。I-JEPA 在预训练中有效捕获低层图像特征，在这些任务中优于视图不变性方法。

不变性方法，如 iBOT [79]，，尽管无需手工数据增强。

少样本 ImageNet-1K。表 2 展示 1% ImageNet 性能，仅用 1 % 标签（每类约 12 或 13 张图）将预训练模型适配为分类器。各方法选择表现更好的微调或线性探测。使用相似编码器时，I-JEPA 以更少预训练轮数超过 MAE。ViT-H/14 的 I-JEPA 匹配 data2vec [8], 预训练的 ViT-L/16，但计算量显著更少（第 7 节）。提高输入分辨率后，超过此前包括预训练额外使用手工增强的联合嵌入方法，如 MSN [4], 、DINO [17], 和 iBOT [79]。

迁移学习。表 3 展示下游图像分类的线性探测。I-JEPA 显著超过同样不用增强的 MAE 和 data2vec，缩小与预训练使用手工增强的最佳视图不变性方法的差距；在 CIFAR100 和 Place205 甚至超过常用 DINO [18] 。

6. 局部预测任务

第 5 节表明，I-JEPA 学到语义图像表征，显著改善 MAE、data2vec 等此前方法的下游分类性能。扩大规模还能缩小与下列方法的差距，甚至超过

使用额外手工增强的视图不变性方法。本节发现，I-JEPA 也学习局部特征，在物体计数、深度预测等低层和密集预测任务中超过这些方法。

表 4 展示各低层任务的线性探测性能。预训练后冻结编码器，在其上训练线性模型，进行 Clevr [43]物体计数和深度预测。相较 DINO、iBOT 等视图不变性方法，I-JEPA 在预训练中有效捕获低层特征，物体计数（Clevr/Count）更好，深度预测（Clevr/Dist）则大幅领先。

7. 扩展性

模型效率。I-JEPA 比此前方法更易扩展。图 5 展示 ImageNet-1K 1% 半监督评估随 GPU 小时数的变化。它无需手工增强，以较少计算取得良好性能。相对直接用像素为目标的 MAE 等重建方法，计算表征目标会产生额外开销（每次迭代约慢 7%）。但 I-JEPA 用约 5×更少的迭代即可收敛，实际仍显著节省计算。相对依靠手工增强创建并处理多视图的 iBOT 等视图不变性方法，I-JEPA 也快得多。巨大 I-JEPA（ViT-H/14）的计算量甚至少于小型 iBOT（ViT-S/16）。

扩大数据量。I-JEPA 也受益于更大预训练数据集。表 5 展示迁移

Pretrain	Arch.	CIFAR100	Place205	INat18	Clevr/Count	Clevr/Dist
IN1k	ViT-H/14	87.5	58.4	47.6	86.7	72.4
IN22k	ViT-H/14	**89.5**	57.8	50.5	**88.6**	**75.0**
IN22k	ViT-G/16	**89.5**	**59.1**	**55.3**	86.7	73.0

Table 5. **Ablating dataset and model size**. Evaluating impact of pre-training dataset size and model size on transfer tasks. I-JEPA benefits from larger more diverse datasets. When increasing the size of the pretraining dataset (IN1k versus IN22k) we see an performance improvement for the ViT-H/14 model. We observe a further performance improvement on semantic tasks by training a larger model ViT-G/16 model on ImageNet-22k. The ViT-H/14 is trained for 300 epochs on IN1k and the equivalent of 900 IN1K epochs on IN22k. The ViT-H/16 is trained for the equivalent of 600 IN1k epochs.

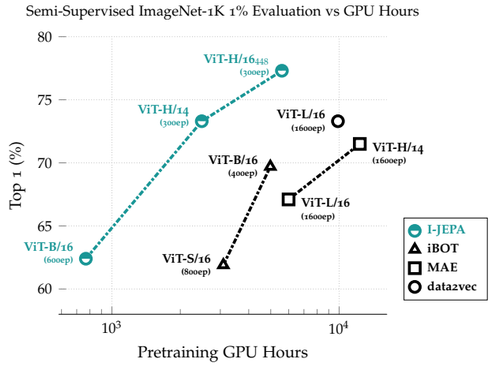


Figure 5. **Scaling**. Semi-supervised evaluation on ImageNet-1K 1% as a function of pretraining GPU hours. I-JEPA requires less compute than previous methods to achieve strong performance. Compared to MAE and data2vec, I-JEPA obtains a significant speedup by requiring fewer pretraining epochs. Compared to iBOT, which relies on hand-crafted data-augmentations, a huge I-JEPA model (ViT-H/14) requires less compute than their smallest model (ViT-S/16).

learning performance on semantic and low level tasks when increasing the size of the pretraining dataset (IN1K versus IN22K). Transfer learning performance on these conceptually different tasks improves when pretraining on a larger more diverse dataset.

Scaling model size. Table 5 also shows that I-JEPA benefit from larger model size when pretraining on IN22K. Pretraining a ViT-G/16 significantly improves the downstream performances on image classification tasks such as Place205 and INat18 compared to a ViT-H/14 model, but does not improve performance on low-level downstream tasks — the ViT-G/16 uses larger input patches, which can be detrimental for the local prediction tasks.

8. Predictor Visualizations

The role of the predictor in I-JEPA is to take the output of the context encoder and, conditioned on positional mask tokens, to predict the representations of a target black at the location specified by the mask tokens. One natural question is whether the predictor conditioned on the positional mask tokens is learning to correctly capture positional uncertainty in the target. To qualitatively investigate this question, we visualize the outputs of the predictor. We use the following visualization approach to enable the research community to independently reproduce our findings. After pretraining, we freeze the context-encoder and predictor weights, and train a decoder following the RCDM framework [13] to map the average-pool of the predictor outputs back to pixel space. Figure 6 shows decoder outputs for various random seeds. Qualities that are common across samples represent information that is contained in the average-pooled predictor representation. The I-JEPA predictor correctly captures positional uncertainty and produces high-level object parts with the correct pose (e.g., back of the bird and top of the car).

9. Ablations

Predicting in representation space. Table 7 compares low-shot performance on 1% ImageNet-1K using a linear probe when the loss is computed in pixel-space versus representation space. We conjecture that a crucial component of I-JEPA is that the loss is computed entirely in representation space, thereby giving the target encoder the ability to produce abstract prediction targets, for which irrelevant pixel-level details are eliminated. From Table 7, it is clear that predicting in pixel-space leads to a significant degradation in the linear probing performance.

Masking strategy. Table 6 compare our `multi-block` masking with other masking strategies such as `rasterized` masking, where the image is split into four large quadrants, and the goal is to use one quadrant as a context to predict the other three quadrants, and the traditional `block` and `random` masking typically used in reconstruction-based methods. In `block` masking, the target is a single image block and the context is the

预训练	架构	CIFAR100	Place205	INat18	Clevr/Count	Clevr/Dist
IN1k	ViT-H/14	87.5	58.4	47.6	86.7	72.4
IN22k	ViT-H/14	**89.5**	57.8	50.5	**88.6**	**75.0**
IN22k	ViT-G/16	**89.5**	**59.1**	**55.3**	86.7	73.0

表 5. 数据集与模型规模消融，评估二者对迁移任务的影响。I-JEPA 受益于更大、更多样的数据。预训练从 IN1k 增至 IN22k，ViT-H/14 性能提高；在 IN22k 上训练更大的 ViT-G/16，语义任务进一步提高。ViT-H/14 在 IN1k 训练 300 轮，在 IN22k 训练相当于 900 个 IN1K 轮；ViT-H/16 训练相当于 600 个 IN1k 轮。

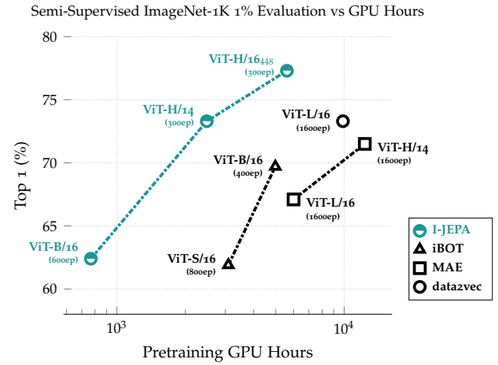


图 5. 扩展性。ImageNet-1K 1% 半监督评估随预训练 GPU 小时数变化。I-JEPA 以更少计算取得良好性能；相比 MAE、data2vec，通过减少预训练轮数明显加速；巨大的 ViT-H/14 甚至比依赖手工增强的 iBOT 最小模型 ViT-S/16 计算量更少。

学习性能，比较预训练数据量增加（IN1K 对 IN22K）时的语义和低层任务。更大、更多样的数据改善这两类概念不同任务的迁移性能。

扩大模型。表 5 表明，在 IN22K 预训练时，I-JEPA 受益于更大模型。ViT-G/16 相对 ViT-H/14 显著提高 Place205、INat18 等分类任务，但不改善低层任务；ViT-G/16 输入图块更大，可能不利于局部预测。

8. 预测器可视化

I-JEPA 预测器接收上下文编码器输出，以位置掩码令牌为条件，预测令牌指定位置的目标块表征。自然的问题是：它是否正确捕获目标中的位置不确定性？为定性研究，我们可视化预测器输出。采用如下可独立复现的方案：预训练后冻结上下文编码器和预测器，按 RCDM 框架 [13] 训练解码器，将预测器输出的平均池化表征映射回像素空间。图 6 展示不同随机种子的解码输出。样本共有的属性代表平均池化预测器表征中包含的信息。I-JEPA 预测器正确捕获位置不确定性，生成姿态正确的高层物体部位（如鸟背和车顶）。

9. 消融

在表征空间预测。表 7 在 1% ImageNet-1K 线性探测中，比较像素空间与表征空间计算损失。我们认为，损失完全在表征空间计算是关键：目标编码器可产生抽象目标，消除无关像素细节。表 7 清楚显示，像素空间预测显著降低线性探测性能。

掩码策略。表 6 比较多块策略和其他策略。象限掩码将图像分为四个大区域，用一个上下文区域预测其余三个；另比较重建方法常用的块状和随机掩码。块状策略以单个图像块为目标，上下文是

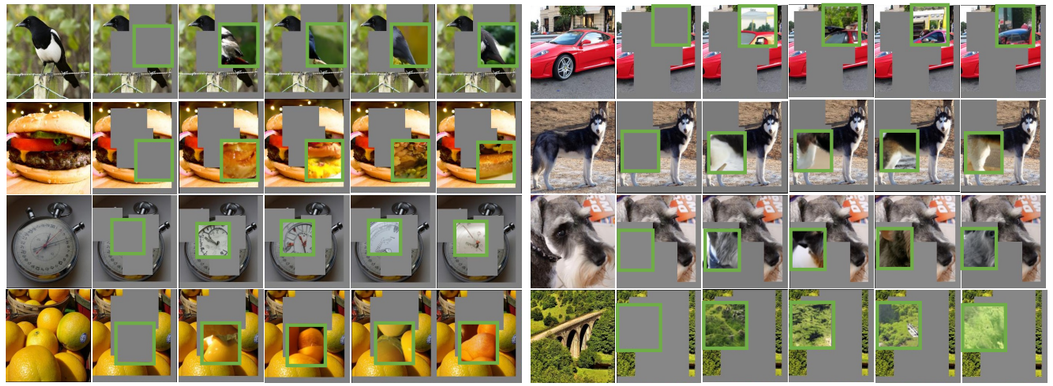


Figure 6. **Visualization of I-JEPA predictor representations**. For each image: first column contains the original image; second column contains the context image, which is processed by a pretrained I-JEPA ViT-H/14 encoder. Green bounding boxes in subsequent columns contain samples from a generative model decoding the output of the pretrained I-JEPA predictor, which is conditioned on positional mask tokens corresponding to the location of the green bounding box. Qualities that are common across samples represent information that contained is in the I-JEPA prediction. The I-JEPA predictor is correctly capturing positional uncertainty and producing high-level object parts with a correct pose (e.g., the back of the bird and top of a car). Qualities that vary across samples represent information that is not contained in the representation. In this case, the I-JEPA predictor discards the precise low-level details as well as background information.

	Targets		**Context**		
Mask	Type	Freq.	Type	Avg. Ratio*	**Top-1**
`multi-block`	Block$(0.15, 0.2)$	4	Block$(0.85, 1.0)$ × Complement	0.25	**54.2**
`rasterized`	Quadrant	3	Complement	0.25	15.5
`block`	Block(0.6)	1	Complement	0.4	20.2
`random`	Random(0.6)	1	Complement	0.4	17.6

**Avg. Ratio is the average number of patches in the context block relative to the total number of patches in the image.*

Table 6. **Ablating masking strategy**. Linear evaluation on ImageNet-1K using only 1% of the available labels after I-JEPA pretraining of a ViT-B/16 for 300 epochs. Comparison of proposed `multi-block` masking strategy. In `rasterized` masking the image is split into four large quadrants; one quadrant is used as a context to predict the other three quadrants. In `block` masking, the target is a single image block and the context is the image complement. In `random` masking, the target is a set of random image patches and the context is the image complement. The proposed `multi-block` masking strategy is helpful for guiding I-JEPA to learn semantic representations.

Targets	**Arch.**	**Epochs**	**Top-1**
Target-Encoder Output	ViT-L/16	500	**66.9**
Pixels	ViT-L/16	800	40.7

Table 7. **Ablating targets**. Linear evaluation on ImageNet-1K using only 1% of the available labels. The semantic level of the I-JEPA representations degrades significantly when the loss is applied in pixel space, rather than representation space, highlighting the importance of the target-encoder during pretraining.

image complement. In `random` masking, the target is a set of random patches and the context is the image complement. Note that there is no overlap between the context and target blocks in all considered strategies. We find `multi-block` masking helpful for guiding I-JEPA to learning semantic representations. Additional ablations on `multi-block` masking can be found in Appendix C.

10. Conclusion

We proposed I-JEPA, a simple and efficient method for learning semantic image representations without relying on hand-crafted data augmentations. We show that by predicting in representation space, I-JEPA converges faster than pixel reconstruction methods and learns representations of a high semantic level. In contrast to view-invariance based methods, I-JEPA highlights a path for learning general representations with joint-embedding architectures, without relying on hand-crafted view augmentations.

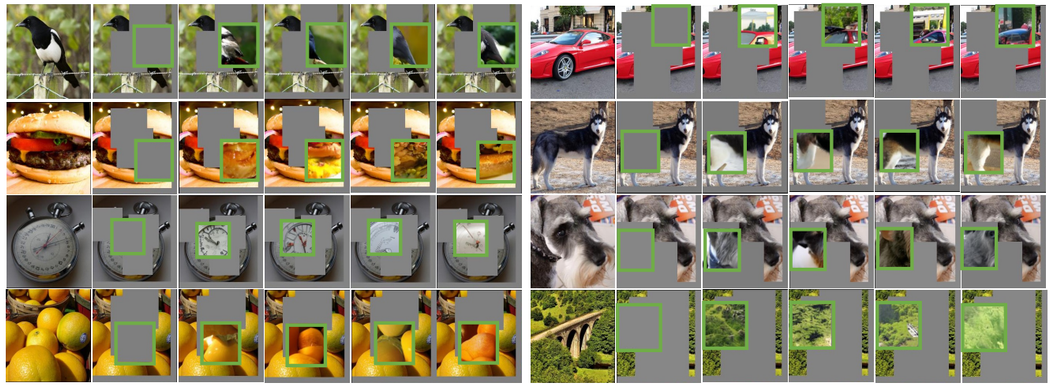


图 6. I-JEPA 预测器表征可视化。首列为原图，第二列为预训练 I-JEPA ViT-H/14 编码器处理的上下文图像。后续列的绿色框是生成模型解码预训练预测器输出的样本；预测器以对应绿色框位置的掩码令牌为条件。样本共有的属性表示预测所包含的信息。预测器正确捕获位置不确定性，生成姿态正确的高层部位（如鸟背、车顶）。样本间变化的属性表示表征未包含的信息：精确低层细节和背景信息被舍弃。

	目标块		**上下文**		
Mask	Type	频率	Type	平均比例*	**Top-1**
多块	块（$0.15, 0.2$）	4	块（$0.85, 1.0$）× 其余区域	0.25	**54.2**
象限	象限	3	其余区域	0.25	15.5
块状	块（0.6）	1	其余区域	0.4	20.2
随机	随机（0.6）	1	其余区域	0.4	17.6

**Avg. Ratio* 为上下文图块数占全部图块数的平均比例。

表 6. 掩码策略消融。ViT-B/16 经 I-JEPA 预训练 300 轮后，仅用 1% ImageNet-1K 标签线性评估。比较多块策略：象限策略将图像分为四块，用一块预测其余三块；块状策略以单个图像块为目标、其余部分为上下文；随机策略以随机图块集合为目标、其余部分为上下文。多块策略有助于引导 I-JEPA 学习语义表征。

目标块	**架构**	**轮数**	**Top-1**
目标编码器输出	ViT-L/16	500	**66.9**
像素	ViT-L/16	800	40.7

表 7. 目标消融，仅用 1% ImageNet-1K 标签线性评估。损失从表征空间改到像素空间后，表征语义层次显著下降，说明预训练目标编码器的重要性。

图像的其余部分。随机掩码以随机图块集合为目标，其余图像为上下文。所有比较策略中，上下文与目标都不重叠。我们发现多块掩码有助于引导 I-JEPA 学习语义表征；其他多块消融见附录 C。

10. 结论

我们提出 I-JEPA：一种无需手工数据增强、简单高效地学习语义图像表征的方法。在表征空间预测，使其比像素重建收敛更快，表征语义层次更高。相对视图不变性方法，I-JEPA 展示了无需手工视图增强、通过联合嵌入架构学习通用表征的路径。

References

[1] Yuki Markus Asano, Christian Rupprecht, and Andrea Vedaldi. Self-labelling via simultaneous clustering and representation learning. *Internatinoal Conference on Learning Representations*, 2020. 1

[2] Mahmoud Assran, Randall Balestriero, Quentin Duval, Florian Bordes, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, and Nicolas Ballas. The hidden uniform cluster prior in self-supervised learning. *International Conference on Learning Representations*, 2023. 1, 13

[3] Mahmoud Assran, Nicolas Ballas, Lluis Castrejon, and Michael Rabbat. Supervision accelerates pre-training in contrastive semi-supervised learning of visual representations. *NeurIPS Workshop on Self-Supervised Learning*, 2020. 13

[4] Mahmoud Assran, Mathilde Caron, Ishan Misra, Piotr Bojanowski, Florian Bordes, Pascal Vincent, Armand Joulin, Michael Rabbat, and Nicolas Ballas. Masked siamese networks for label-efficient learning. *European Conference on Computer Vision*, 2022. 1, 2, 3, 5, 6, 12, 13, 16, 17

[5] Mahmoud Assran, Mathilde Caron, Ishan Misra, Piotr Bojanowski, Armand Joulin, Nicolas Ballas, and Michael Rabbat. Semi-supervised learning of visual features by non-parametrically predicting view assignments with support samples. *IEEE/CVF International Conference on Computer Vision*, 2021. 3, 13

[6] Philip Bachman, R Devon Hjelm, and William Buchwalter. Learning representations by maximizing mutual information across views. *Advances in neural information processing systems*, 32, 2019. 13

[7] Alexei Baevski, Arun Babu, Wei-Ning Hsu, and Michael Auli. Efficient self-supervised learning with contextualized target representations for vision, speech and language. *arXiv preprint arXiv:2212.07525*, 2022. 5

[8] Alexei Baevski, Wei-Ning Hsu, Qiantong Xu, Arun Babu, Jiatao Gu, and Michael Auli. Data2vec: A general framework for self-supervised learning in speech, vision and language. *arXiv preprint arXiv:2202.03555*, 2022. 1, 3, 4, 5, 6, 13

[9] Hangbo Bao, Li Dong, and Furu Wei. Beit: Bert pre-training of image transformers. *arXiv preprint arXiv:2106.08254*, 2021. 1, 3, 4, 13

[10] Adrien Bardes, Jean Ponce, and Yann LeCun. Vicreg: Variance-invariance-covariance regularization for self-supervised learning. *arXiv preprint arXiv:2105.04906*, 2021. 1, 3, 13

[11] Adrien Bardes, Jean Ponce, and Yann LeCun. Vicregl: Self-supervised learning of local visual features. *arXiv preprint arXiv:2210.01571*, 2022. 1, 13

[12] Florian Bordes, Randall Balestriero, Quentin Garrido, Adrien Bardes, and Pascal Vincent. Guillotine regularization: Improving deep networks generalization by removing their head. *arXiv preprint arXiv:2206.13378*, 2022. 13

[13] Florian Bordes, Randall Balestriero, and Pascal Vincent. High fidelity visualization of what your self-supervised representation knows about. *Transactions on Machine Learning Research*, 2022. 7, 16

[14] John Bridle, Anthony Heading, and David MacKay. Unsupervised classifiers, mutual information and'phantom targets. *Advances in neural information processing systems*, 4, 1991. 13

[15] Jane Bromley, James W Bentz, Léon Bottou, Isabelle Guyon, Yann LeCun, Cliff Moore, Eduard Säckinger, and Roopak Shah. Signature verification using a "siamese" time delay neural network. *International Journal of Pattern Recognition and Artificial Intelligence*, 7(04):669–688, 1993. 1, 3

[16] Zhaowei Cai, Avinash Ravichandran, Paolo Favaro, Manchen Wang, Davide Modolo, Rahul Bhotika, Zhuowen Tu, and Stefano Soatto. Semi-supervised vision transformers at scale. *arXiv preprint arXiv:2208.05688*, 2022. 13

[17] Mathilde Caron, Ishan Misra, Julien Mairal, Priya Goyal, Piotr Bojanowski, and Armand Joulin. Unsupervised learning of visual features by contrasting cluster assignments. *arXiv preprint arXiv:2006.09882*, 2020. 1, 6

[18] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. *arXiv preprint arXiv:2104.14294*, 2021. 1, 3, 4, 5, 6, 12, 13

[19] Mark Chen, Alec Radford, Rewon Child, Jeffrey Wu, Heewoo Jun, David Luan, and Ilya Sutskever. Generative pretraining from pixels. In *International Conference on Machine Learning*, pages 1691–1703. PMLR, 2020. 13

[20] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. *preprint arXiv:2002.05709*, 2020. 1, 2, 13

[21] Ting Chen, Simon Kornblith, Kevin Swersky, Mohammad Norouzi, and Geoffrey Hinton. Big self-supervised models are strong semi-supervised learners. *arXiv preprint arXiv:2006.10029*, 2020. 5

[22] Xiaokang Chen, Mingyu Ding, Xiaodi Wang, Ying Xin, Shentong Mo, Yunhao Wang, Shumin Han, Ping Luo, Gang Zeng, and Jingdong Wang. Context autoencoder for self-supervised representation learning. *arXiv preprint arXiv:2202.03026*, 2022. 5

[23] Xinlei Chen, Haoqi Fan, Ross Girshick, and Kaiming He. Improved baselines with momentum contrastive learning. *arXiv preprint arXiv:2003.04297*, 2020. 12, 13

[24] Xinlei Chen and Kaiming He. Exploring simple siamese representation learning. *arXiv preprint arXiv:2011.10566*, 2020. 1, 3, 13

[25] Xinlei Chen, Saining Xie, and Kaiming He. An empirical study of training self-supervised vision transformers. *arXiv preprint arXiv:2104.02057*, 2021. 4

[26] Yubei Chen, Adrien Bardes, Zengyi Li, and Yann LeCun. Intra-instance vicreg: Bag of self-supervised image patch embedding. *arXiv preprint arXiv:2206.08954*, 2022. 13

[27] Navneet Dalal and Bill Triggs. Histograms of oriented gradients for human detection. In *2005 IEEE computer society conference on computer vision and pattern recognition (CVPR'05)*, volume 1, pages 886–893. Ieee, 2005. 4

[28] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018. 1

参考文献

[1] Yuki Markus Asano, Christian Rupprecht, and Andrea Vedaldi. Self-labelling via simultaneous clustering and representation learning. Internatinoal Conference on Learning Representations, 2020. 1[2] Mahmoud Assran, Randall Balestriero, Quentin Duval, Flo- rian Bordes, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, and Nicolas Ballas. The hidden uniform cluster prior in self-supervised learning. International Con- ference on Learning Representations, 2023. 1, 13[3] Mahmoud Assran, Nicolas Ballas, Lluis Castrejon, and Michael Rabbat. Supervision accelerates pre-training in con- trastive semi-supervised learning of visual representations. NeurIPS Workshop on Self-Supervised Learning, 2020. 13[4] Mahmoud Assran, Mathilde Caron, Ishan Misra, Piotr Bo- janowski, Florian Bordes, Pascal Vincent, Armand Joulin, Michael Rabbat, and Nicolas Ballas. Masked siamese net- works for label-efficient learning. European Conference on Computer Vision, 2022. 1, 2, 3, 5, 6, 12, 13, 16, 17[5] Mahmoud Assran, Mathilde Caron, Ishan Misra, Piotr Bo- janowski, Armand Joulin, Nicolas Ballas, and Michael Rab- bat. Semi-supervised learning of visual features by non- parametrically predicting view assignments with support samples. IEEE/CVF International Conference on Computer Vision, 2021. 3, 13[6] Philip Bachman, R Devon Hjelm, and William Buchwalter. Learning representations by maximizing mutual information across views. Advances in neural information processing systems, 32, 2019. 13[7] Alexei Baevski, Arun Babu, Wei-Ning Hsu, and Michael Auli. Efficient self-supervised learning with contextualized target representations for vision, speech and language. arXiv preprint arXiv:2212.07525, 2022. 5[8] Alexei Baevski, Wei-Ning Hsu, Qiantong Xu, Arun Babu, Ji- atao Gu, and Michael Auli. Data2vec: A general framework for self-supervised learning in speech, vision and language. arXiv preprint arXiv:2202.03555, 2022. 1, 3, 4, 5, 6, 13[9] Hangbo Bao, Li Dong, and Furu Wei. Beit: Bert pre-training of image transformers. arXiv preprint arXiv:2106 08254, 2021 1, 3, 4, 13 . . [10]Adrien Bardes, Jean Ponce, and Yann LeCun. Vi- creg: Variance-invariance-covariance regularization for self-supervised learning. arXiv preprint arXiv:2105.04906, 2021. 1, 3, 13[11] Adrien Bardes, Jean Ponce, and Yann LeCun. Vicregl: Self- supervised learning of local visual features. arXiv preprint arXiv:2210.01571, 2022. 1, 13[12]Florian Bordes, Randall Balestriero, Quentin Garrido, Adrien Bardes, and Pascal Vincent. Guillotine regulariza- tion: Improving deep networks generalization by removing their head. arXiv preprint arXiv:2206.13378, 2022. 13[13] Florian Bordes, Randall Balestriero, and Pascal Vincent. High fidelity visualization of what your self-supervised rep- resentation knows about. Transactions on Machine Learning Research, 2022. 7, 16[14] John Bridle, Anthony Heading, and David MacKay. Un- supervised classifiers, mutual information and' phantom tar-

gets. Advances in neural information processing systems, 4, 1991 13. [15] Jane Bromley, James W Bentz, L´eon Bottou, Isabelle Guyon, Yann LeCun, Cliff Moore, Eduard S¨ackinger, and Roopak Shah. Signature verification using a "siamese" time delay neural network. International Journal of Pattern Recognition and Artificial Intelligence, 7(04):669–688, 1993. 1, 3[16]Zhaowei Cai, Avinash Ravichandran, Paolo Favaro, Manchen Wang, Davide Modolo, Rahul Bhotika, Zhuowen Tu, and Stefano Soatto. Semi-supervised vision transformers at scale. arXiv preprint arXiv:2208.05688, 2022. 13[17] Mathilde Caron, Ishan Misra, Julien Mairal, Priya Goyal, Pi- otr Bojanowski, and Armand Joulin. Unsupervised learning of visual features by contrasting cluster assignments. arXiv preprint arXiv:2006.09882, 2020. 1, 6[18] Mathilde Caron, Hugo Touvron, Ishan Misra, Herv´e J´egou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerg- ing properties in self-supervised vision transformers. arXiv preprint arXiv:2104.14294, 2021. 1, 3, 4, 5, 6, 12, 13[19] Mark Chen, Alec Radford, Rewon Child, Jeffrey Wu, Hee- woo Jun, David Luan, and Ilya Sutskever. Generative pre- training from pixels. In International Conference on Ma- chine Learning, pages 1691–1703. PMLR, 2020. 13[20] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Ge- offrey Hinton. A simple framework for contrastive learning of visual representations. preprint arXiv:2002.05709, 2020. 1, 2, 13[21] Ting Chen, Simon Kornblith, Kevin Swersky, Mohammad Norouzi, and Geoffrey Hinton. Big self-supervised mod- els are strong semi-supervised learners. arXiv preprint arXiv:2006.10029, 2020. 5[22] Xiaokang Chen, Mingyu Ding, Xiaodi Wang, Ying Xin, Shentong Mo, Yunhao Wang, Shumin Han, Ping Luo, Gang Zeng, and Jingdong Wang. Context autoencoder for self-supervised representation learning. arXiv preprint arXiv:2202.03026, 2022. 5[23] Xinlei Chen, Haoqi Fan, Ross Girshick, and Kaiming He. Improved baselines with momentum contrastive learning. arXiv preprint arXiv:2003.04297, 2020. 12, 13[24] Xinlei Chen and Kaiming He. Exploring simple siamese representation learning. arXiv preprint arXiv:2011 10566, 2020 1, 3, 13 . . [25] Xinlei Chen, Saining Xie, and Kaiming He. An empirical study of training self-supervised vision transformers. arXiv preprint arXiv:2104.02057, 2021. 4[26] Yubei Chen, Adrien Bardes, Zengyi Li, and Yann LeCun. Intra-instance vicreg: Bag of self-supervised image patch embedding. arXiv preprint arXiv:2206.08954, 2022. 13[27] Navneet Dalal and Bill Triggs. Histograms of oriented gra- dients for human detection. In 2005 IEEE computer soci- ety conference on computer vision and pattern recognition (CVPR' 05), volume 1, pages 886–893. Ieee, 2005. 4[28] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. arXiv preprint arXiv:1810.04805, 2018. 1

[29] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020. 3, 4, 12, 13

[30] Alaaeldin El-Nouby, Gautier Izacard, Hugo Touvron, Ivan Laptev, Hervé Jegou, and Edouard Grave. Are large-scale datasets necessary for self-supervised pre-training? *arXiv preprint arXiv:2112.10740*, 2021. 13

[31] Karl Friston. A theory of cortical responses. *Philosophical transactions of the Royal Society B: Biological sciences*, 360(1456):815–836, 2005. 1

[32] Spyros Gidaris, Andrei Bursuc, Nikos Komodakis, Patrick Pérez, and Matthieu Cord. Learning representations by predicting bags of visual words. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6928–6938, 2020. 13

[33] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep learning*. MIT press, 2016. 13

[34] Priya Goyal, Quentin Duval, Jeremy Reizenstein, Matthew Leavitt, Min Xu, Benjamin Lefaudeux, Mannat Singh, Vinicius Reis, Mathilde Caron, Piotr Bojanowski, Armand Joulin, and Ishan Misra. Vissl. https://github.com/facebookresearch/vissl, 2021. 12

[35] Jean-Bastien Grill, Florian Strub, Florent Altché, Corentin Tallec, Pierre H Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pires, Zhaohan Daniel Guo, Mohammad Gheshlaghi Azar, et al. Bootstrap your own latent: A new approach to self-supervised learning. *arXiv preprint arXiv:2006.07733*, 2020. 1, 3, 5, 12, 13

[36] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. 1, 2, 3, 4, 5, 6, 12, 13, 15, 16

[37] Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross Girshick. Momentum contrast for unsupervised visual representation learning. *arXiv preprint arXiv:1911.05722*, 2019. 1, 3, 12, 13

[38] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 770–778, 2016. 3

[39] Olivier Henaff. Data-efficient image recognition with contrastive predictive coding. In *International conference on machine learning*, pages 4182–4192. PMLR, 2020. 13

[40] R Devon Hjelm, Alex Fedorov, Samuel Lavoie-Marchildon, Karan Grewal, Phil Bachman, Adam Trischler, and Yoshua Bengio. Learning deep representations by mutual information estimation and maximization. *arXiv preprint arXiv:1808.06670*, 2018. 13

[41] Weihua Hu, Takeru Miyato, Seiya Tokui, Eiichi Matsumoto, and Masashi Sugiyama. Learning discrete representations via information maximizing self-augmented training. In *International conference on machine learning*, pages 1558–1567. PMLR, 2017. 13

[42] Justin Johnson, Bharath Hariharan, Laurens Van Der Maaten, Li Fei-Fei, C Lawrence Zitnick, and Ross Girshick. Clevr: A diagnostic dataset for compositional language and elementary visual reasoning. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2901–2910, 2017. 12

[43] Justin Johnson, Bharath Hariharan, Laurens van der Maaten, Li Fei-Fei, C Lawrence Zitnick, and Ross Girshick. Clevr: A diagnostic dataset for compositional language and elementary visual reasoning. In *CVPR*, 2017. 6

[44] Andreas Krause, Pietro Perona, and Ryan Gomes. Discriminative clustering by regularized information maximization. *Advances in neural information processing systems*, 23, 2010. 13

[45] Alex Krizhevsky, Geoffrey Hinton, et al. Learning multiple layers of features from tiny images. 2009. 12

[46] Gustav Larsson, Michael Maire, and Gregory Shakhnarovich. Learning representations for automatic colorization. 2016. 4

[47] Gustav Larsson, Michael Maire, and Gregory Shakhnarovich. Colorization as a proxy task for visual understanding. 2017. 4

[48] Yann LeCun. A path towards autonomous machine intelligence version 0.9. 2, 2022-06-27. 2022. 2, 3

[49] Yann LeCun, Sumit Chopra, Raia Hadsell, M Ranzato, and Fujie Huang. A tutorial on energy-based learning. *Predicting structured data*, 1(0), 2006. 2

[50] Ralph Linsker. Self-organization in a perceptual network. *Computer*, 21(3):105–117, 1988. 13

[51] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017. 12

[52] Yi Ma, Doris Tsao, and Heung-Yeung Shum. On the principles of parsimony and self-consistency for the emergence of intelligence. *Frontiers of Information Technology & Electronic Engineering*, pages 1–26, 2022. 13

[53] Ishan Misra and Laurens van der Maaten. Self-supervised learning of pretext-invariant representations. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 6707–6717, 2020. 13

[54] Jovana Mitrovic, Brian McWilliams, Jacob Walker, Lars Buesing, and Charles Blundell. Representation learning via invariant causal mechanisms. *International Conference on Learning Representations*, 2021. 13

[55] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018. 13

[56] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. Pytorch: An imperative style, high-performance deep learning library. *Advances in neural information processing systems*, 32, 2019. 12

[57] Deepak Pathak, Philipp Krahenbuhl, Jeff Donahue, Trevor Darrell, and Alexei A Efros. Context encoders: Feature learning by inpainting. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2536–2544, 2016. 1, 4

[58] Aditya Ramesh, Mikhail Pavlov, Gabriel Goh, Scott Gray, Chelsea Voss, Alec Radford, Mark Chen, and Ilya Sutskever.

[29] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. arXiv preprint arXiv:2010.11929, 2020. 3, 4, 12, 13[30] Alaaeldin El-Nouby, Gautier Izacard, Hugo Touvron, Ivan Laptev, Herv´e Jegou, and Edouard Grave. Are large-scale datasets necessary for self-supervised pre-training? arXiv preprint arXiv:2112.10740, 2021. 13[31] Karl Friston. A theory of cortical responses. Philosophi- cal transactions of the Royal Society B: Biological sciences, 360(1456):815–836, 2005. 1[32] Spyros Gidaris, Andrei Bursuc, Nikos Komodakis, Patrick P´erez, and Matthieu Cord. Learning representations by predicting bags of visual words. I n Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 6928–6938, 2020. 13[33] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep learning. MIT press, 2016. 13[34] Priya Goyal, Quentin Duval, Jeremy Reizenstein, Matthew Leavitt, Min Xu, Benjamin Lefaudeux, Mannat Singh, Vinicius Reis, Mathilde Caron, Piotr Bojanowski, Armand Joulin, and Ishan Misra. Vissl. https://github.com/ facebookresearch/vissl, 2021. 12[35] Jean-Bastien Grill, Florian Strub, Florent Altch´e, Corentin Tallec, Pierre H Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pires, Zhaohan Daniel Guo, Mohammad Gheshlaghi Azar, et al. Bootstrap your own latent: A new approach to self-supervised learning. arXiv preprint arXiv:2006.07733, 2020. 1, 3, 5, 12, 13[36] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Doll´ar, and Ross Girshick. Masked autoencoders are scalable vision learners. IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022. 1, 2, 3, 4, 5, 6, 12, 13, 15, 16[37] Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross Girshick. Momentum contrast for unsupervised visual repre- sentation learning. arXiv preprint arXiv:1911.05722, 2019. 1, 3, 12, 13[38] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In Proceed- ings of the IEEE Conference on Computer Vision and Pattern Recognition, pages 770–778, 2016. 3[39] Olivier Henaff. Data-efficient image recognition with con- trastive predictive coding. In International conference on machine learning, pages 4182–4192. PMLR, 2020. 13[40] R Devon Hjelm, Alex Fedorov, Samuel Lavoie-Marchildon, Karan Grewal, Phil Bachman, Adam Trischler, and Yoshua Bengio. Learning deep representations by mutual in- formation estimation and maximization. arXiv preprint arXiv:1808.06670, 2018. 13[41] Weihua Hu, Takeru Miyato, Seiya Tokui, Eiichi Matsumoto, and Masashi Sugiyama. Learning discrete representations via information maximizing self-augmented training. In In- ternational conference on machine learning, pages 1558– 1567. PMLR, 2017. 13[42]Justin Johnson, Bharath Hariharan, Laurens Van Der Maaten, Li Fei-Fei, C Lawrence Zitnick, and Ross Girshick. Clevr: A diagnostic dataset for compositional language and elementary visual reasoning. In Proceedings of the IEEE conference on computer vision and pattern recognition, pages 2901–2910, 2017. 12[43] Justin Johnson, Bharath Hariharan, Laurens van der Maaten, Li Fei-Fei, C Lawrence Zitnick, and Ross Girshick. Clevr: A diagnostic dataset for compositional language and elemen- tary visual reasoning. In CVPR, 2017. 6[44] Andreas Krause, Pietro Perona, and Ryan Gomes. Dis- criminative clustering by regularized information maximiza- tion. Advances in neural information processing systems, 23, 2010 13. [45] Alex Krizhevsky, Geoffrey Hinton, et al. Learning multiple layers of features from tiny images. 2009. 12[46] Gustav Larsson, Michael Maire, and Gregory Shakhnarovich. Learning representations for automatic colorization. 2016. 4[47] Gustav Larsson, Michael Maire, and Gregory Shakhnarovich. Colorization as a proxy task for visual understanding. 2017. 4[48] Yann LeCun. A path towards autonomous machine intelli- gence version 0.9. 2, 2022-06-27. 2022. 2, 3[49] Yann LeCun, Sumit Chopra, Raia Hadsell, M Ranzato, and Fujie Huang. A tutorial on energy-based learning. Predicting structured data, 1(0), 2006. 2 [50] Ralph Linsker. Self-organization in a perceptual network. Computer, 21(3):105–117, 1988. 13[51] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. arXiv preprint arXiv:1711.05101, 2017. 12[52] Yi Ma, Doris Tsao, and Heung-Yeung Shum. On the prin- ciples of parsimony and self-consistency for the emergence of intelligence. Frontiers of Information Technology & Elec- tronic Engineering, pages 1–26, 2022. 13[53] Ishan Misra and Laurens van der Maaten. Self-supervised learning of pretext-invariant representations. In Proceed- ings of the IEEE Conference on Computer Vision and Pattern Recognition, pages 6707–6717, 2020. 13[54] Jovana Mitrovic, Brian McWilliams, Jacob Walker, Lars Buesing, and Charles Blundell. Representation learning via invariant causal mechanisms. International Conference on Learning Representations, 2021. 13[55] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Repre- sentation learning with contrastive predictive coding. arXiv preprint arXiv:1807.03748, 2018. 13[56] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, Ja mes Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. Pytorch: An im- perative style, high-performance deep learning library. Ad- vances in neural information processing systems, 32, 2019. 12 [57] Deepak Pathak, Philipp Krahenbuhl, Jeff Donahue, Trevor Darrell, and Alexei A Efros. Context encoders: Feature learning by inpainting. In Proceedings of the IEEE con- ference on computer vision and pattern recognition, pages 2536–2544, 2016. 1, 4[58] Aditya Ramesh, Mikhail Pavlov, Gabriel Goh, Scott Gray, Chelsea Voss, Alec Radford, Mark Chen, and Ilya Sutskever.

Zero-shot text-to-image generation. In *International Conference on Machine Learning*, pages 8821–8831. PMLR, 2021. 4

[59] Rajesh PN Rao and Dana H Ballard. Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects. *Nature neuroscience*, 2(1):79–87, 1999. 1

[60] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, Alexander C. Berg, and Li Fei-Fei. Imagenet large scale visual recognition challenge. *International Journal of Computer Vision*, 115(3):211–252, 2015. 5, 12

[61] Antti Tarvainen and Harri Valpola. Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results. *arXiv preprint arXiv:1703.01780*, 2017. 12

[62] Yuandong Tian, Xinlei Chen, and Surya Ganguli. Understanding self-supervised learning dynamics without contrastive pairs. In *International Conference on Machine Learning*, pages 10268–10278. PMLR, 2021. 13

[63] Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles, and Hervé Jégou. Training data-efficient image transformers & distillation through attention. In *International Conference on Machine Learning*, pages 10347–10357. PMLR, 2021. 3

[64] Michael Tschannen, Josip Djolonga, Paul K Rubenstein, Sylvain Gelly, and Mario Lucic. On mutual information maximization for representation learning. *arXiv preprint arXiv:1907.13625*, 2019. 13

[65] Grant Van Horn, Oisin Mac Aodha, Yang Song, Yin Cui, Chen Sun, Alex Shepard, Hartwig Adam, Pietro Perona, and Serge Belongie. The inaturalist species classification and detection dataset. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 8769–8778, 2018. 12

[66] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in neural information processing systems*, pages 5998–6008, 2017. 3

[67] Pascal Vincent, Hugo Larochelle, Isabelle Lajoie, Yoshua Bengio, Pierre-Antoine Manzagol, and Léon Bottou. Stacked denoising autoencoders: Learning useful representations in a deep network with a local denoising criterion. *Journal of machine learning research*, 11(12), 2010. 1, 4, 13

[68] Chen Wei, Haoqi Fan, Saining Xie, Chao-Yuan Wu, Alan Yuille, and Christoph Feichtenhofer. Masked feature prediction for self-supervised visual pre-training. *arXiv preprint arXiv:2112.09133*, 2021. 1, 13

[69] Zhirong Wu, Yuanjun Xiong, Stella X Yu, and Dahua Lin. Unsupervised feature learning via non-parametric instance discrimination. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3733–3742, 2018. 13

[70] Qizhe Xie, Zihang Dai, Eduard Hovy, Minh-Thang Luong, and Quoc V Le. Unsupervised data augmentation. *arXiv preprint arXiv:1904.12848*, 2019. 13

[71] Zhenda Xie, Zheng Zhang, Yue Cao, Yutong Lin, Jianmin Bao, Zhuliang Yao, Qi Dai, and Han Hu. Simmim: A simple framework for masked image modeling. *arXiv preprint arXiv:2111.09886*, 2021. 1, 4

[72] Yang You, Igor Gitman, and Boris Ginsburg. Large batch training of convolutional networks, 2017. 12

[73] Sangdoo Yun, Dongyoon Han, Seong Joon Oh, Sanghyuk Chun, Junsuk Choe, and Youngjoon Yoo. Cutmix: Regularization strategy to train strong classifiers with localizable features. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 6023–6032, 2019. 16

[74] Jure Zbontar, Li Jing, Ishan Misra, Yann LeCun, and Stéphane Deny. Barlow twins: Self-supervised learning via redundancy reduction. *arXiv preprint arXiv:2103.03230*, 2021. 1, 3, 13

[75] Xiaohua Zhai, Joan Puigcerver, Alexander Kolesnikov, Pierre Ruyssen, Carlos Riquelme, Mario Lucic, Josip Djolonga, Andre Susano Pinto, Maxim Neumann, Alexey Dosovitskiy, Lucas Beyer, Olivier Bachem, Michael Tschannen, Marcin Michalski, Olivier Bousquet, Sylvain Gelly, and Neil Houlsby. A large-scale study of representation learning with the visual task adaptation benchmark, 2019. 12

[76] Hongyi Zhang, Moustapha Cisse, Yann N Dauphin, and David Lopez-Paz. mixup: Beyond empirical risk minimization. *Internatinoal Conference on Learning Representations*, 2018. 16

[77] Richard Zhang, Phillip Isola, and Alexei A Efros. Colorful image colorization. 2016. 4

[78] Bolei Zhou, Agata Lapedriza, Jianxiong Xiao, Antonio Torralba, and Aude Oliva. Learning deep features for scene recognition using places database. *Advances in neural information processing systems*, 27, 2014. 12

[79] Jinghao Zhou, Chen Wei, Huiyu Wang, Wei Shen, Cihang Xie, Alan Yuille, and Tao Kong. Ibot: Image bert pre-training with online tokenizer. *International Conference on Learning Representations*, 2022. 2, 4, 5, 6, 12, 13

Zero-shot text-to-image generation. In International Confer- ence on Machine Learning, pages 8821–8831. PMLR, 2021. 4[59] Rajesh PN Rao and Dana H Ballard. Predictive coding in the visual cortex: a functional interpretation of so me extra-classical receptive-field effects. Nature neuroscience, 2(1):79–87, 1999. 1[60] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, San- jeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, Alexander C. Berg, and Li Fei-Fei. Imagenet large scale visual recognition challenge. International Journal of Computer Vision, 115 252, 2015 3 211 5, 12 ( ): – . [61] Antti Tarvainen and Harri Valpola. Mean teachers are bet- ter role models: Weight-averaged consistency targets im- prove semi-supervised deep learning results. arXiv preprint arXiv:1703.01780, 2017. 12[62] Yuandong Tian, Xinlei Chen, and Surya Ganguli. Un- derstanding self-supervised learning dynamics without con- trastive pairs. In International Conference on Machine Learning, pages 10268–10278. PMLR, 2021. 13[63] Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles, and Herv´e J´egou. Training data-efficient image transformers & distillation through at- tention. In International Conference on Machine Learning, pages 10347–10357. PMLR, 2021. 3[64] Michael Tschannen, Josip Djolonga, Paul K Rubenstein, Sylvain Gelly, and Mario Lucic. On mutual information maximization for representation learning. arXiv preprint arXiv:1907.13625, 2019. 13[65] Grant Van Horn, Oisin Mac Aodha, Yang Song, Yin Cui, Chen Sun, Alex Shepard, Hartwig Adam, Pietro Perona, and Serge Belongie. The inaturalist species classification and de- tection dataset. In Proceedings of the IEEE conference on computer vision and pattern recognition, pages 8769 8778, 2018 12 – . [66] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszko- reit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In Advances in neural information processing systems, pages 5998–6008, 2017. 3[67] Pascal Vincent, Hugo Larochelle, Isabelle Lajoie, Yoshua Bengio, Pierre-Antoine Manzagol, and L´eon Bottou. Stacked denoising autoencoders: Learning useful represen- tations in a deep network with a local denoising criterion. Journal of machine learning research, 11(12), 2010. 1, 4, 13[68] Chen Wei, Haoqi Fan, Saining Xie, Chao-Yuan Wu, Alan Yuille, and Christoph Feichtenhofer. Masked feature predic- tion for self-supervised visual pre-training. arXiv preprint arXiv:2112.09133, 2021. 1, 13[69] Zhirong Wu, Yuanjun Xiong, Stella X Yu, and Dahua Lin. Unsupervised feature learning via non-parametric instance discrimination. In Proceedings of the IEEE conference on computer vision and pattern recognition, pages 3733 3742, 2018 13 – . [70] Qizhe Xie, Zihang Dai, Eduard Hovy, Minh-Thang Luong, and Quoc V Le. Unsupervised data augmentation. arXiv preprint arXiv:1904.12848, 2019. 13

[71] Zhenda Xie, Zheng Zhang, Yue Cao, Yutong Lin, Jianmin Bao, Zhuliang Yao, Qi Dai, and Han Hu. Simmim: A sim- ple framework for masked image modeling. arXiv preprint arXiv:2111.09886, 2021. 1, 4[72] Yang You, Igor Gitman, and Boris Ginsburg. Large batch training of convolutional networks, 2017. 12[73] Sangdoo Yun, Dongyoon Han, Seong Joon Oh, Sanghyuk Chun, Junsuk Choe, and Youngjoon Yoo. Cutmix: Regular- ization strategy to train strong classifiers with localizable fea- tures. In Proceedings of the IEEE/CVF International Con- ference on Computer Vision, pages 6023–6032, 2019. 16[74] Jure Zbontar, Li Jing, Ishan Misra, Yann LeCun, and St´ephane Deny. Barlow twins: Self-supervised learning via redundancy reduction. arXiv preprint arXiv:2103 03230, 2021 1, 3, 13 . . [75]Xiaohua Zhai, Joan Puigcerver, Alexander Kolesnikov, Pierre Ruyssen, Carlos Riquelme, Mario Lucic, Josip Djo- longa, Andre Susano Pinto, Maxim Neumann, Alexey Doso- vitskiy, Lucas Beyer, Olivier Bachem, Michael Tschannen, Marcin Michalski, Olivier Bousquet, Sylvain Gelly, and Neil Houlsby. A large-scale study of representation learning with the visual task adaptation benchmark, 2019. 12[76] Hongyi Zhang, Moustapha Cisse, Yann N Dauphin, and David Lopez-Paz. mixup: Beyond empirical risk minimiza- tion. Internatinoal Conference on Learning Representations, 2018. 16[77] Richard Zhang, Phillip Isola, and Alexei A Efros. Colorful image colorization. 2016. 4 [78] Bolei Zhou, Agata Lapedriza, Jianxiong Xiao, Antonio Tor- ralba, and Aude Oliva. Learning deep features for scene recognition using places database. Advances in neural in- formation processing systems, 27, 2014. 12[79] Jinghao Zhou, Chen Wei, Huiyu Wang, Wei Shen, Cihang Xie, Alan Yuille, and Tao Kong. Ibot: Image bert pre- training with online tokenizer. International Conference on Learning Representations, 2022. 2, 4, 5, 6, 12, 13

A. Implementation Details

A.1. Pretraining

Architectures. For I-JEPA pretraining, we use Vision Transformer [29] (ViT) architectures for the context-encoder, target-encoder, and the predictor. While the context-encoders and target-encoders correspond to standard ViT architectures, the predictor is designed as a light-weight (narrow) ViT architecture. Specifically, we fix the embedding dimension of the predictor to 384, while keeping the number of self-attention heads equal to that of the backbone context-encoder. For the smaller ViT-B/16 context-encoder, we set the depth of the predictor to 6. For ViT-L/16, ViT-H/16, and ViT-H/14 context-encoders, we set the depth of the predictor to 12. Finally, the ViT-G/16 uses a predictor of depth 16. I-JEPA is pretrained without a `[cls]` token. We use the target-encoder for evaluation and average pool its output to produce a global image representation.

Optimization. We use AdamW [51] to optimize the context-encoder and predictor weights. Our default batch-size is 2048, and the learning rate is linearly increased from 10^{-4} to 10^{-3} during the first 15 epochs of pretraining, and decayed to 10^{-6} following a cosine schedule thereafter. Following [4, 18], the weight-decay is linearly increased from 0.04 to 0.4 throughout pretraining. The target-encoder weights are identical to the context-encoder weights at initialization, and updated via an exponential moving average thereafter [4, 18, 23, 35, 37, 61]. We use a momentum value of 0.996, and linearly increase this value to 1.0 throughout pretraining, following [4, 18].

Masking. By default, we sample 4 possibly overlapping target blocks masks with random scale in the range $(015, 0.2)$ and aspect ratio in the range $(0.75, 1.5)$. We sample 1 context block mask with random scale in the range $(0.85, 1.0)$ and unit aspect ratio. We subsequently eliminate any regions in the context block mask that overlap with any of the 4 target block masks. The context-block mask and target-block masks are sampled independently for each image in the mini-batch. To ensure efficient batch processing, we restrict the size of all context masks on a co-located GPU to be identical. Similarly, we restrict the size of all target masks on a co-located GPU to be identical. The mask-sampler is efficiently implemented in only a few lines of code in PyTorch [56] using a batch-collator function, which runs in the data loader processes. In short, in each iteration, the data loader returns a mini-batch of images and a set of context and target masks for each image, identifying the patch indices to keep for the context and target views.

A.2. Downstream Tasks

Linear evaluation. When evaluating methods such as iBOT [79], DINO [18] or MAE [36], which leverage Vision Transformers [29] with an additional `[cls]` token, we use the default configurations of VISSL [34] to evaluate all the models on iNaturalist18 [65], CIFAR100 [45], Clevr/Count [42, 75], Clevr/Dist [42, 75], and Places205 [78]. We freeze the encoder and return the best number among the following representations: 1) the `[cls]` token representation of the last layer, 2) the concatenation of the last 4 layers of the `[cls]` token. For each representation, we try two different heads: 1) a linear head, or 2) a linear head preceded by a batch normalization, and return the best number. We use the default data augmentations of VISSL [34]: random resize cropping and horizontal flipping, with the exception of Clevr/Count and Clevr/Dist, where we only use center crop and horizontal flipping, as random cropping interferes with the capability of counting objects and estimating distance, removing critical objects from the scene. For CIFAR100, we resize the images to 224×224 pixels, so as to keep the number of patches equal to that used during pretraining.

Because our I-JEPA implementation uses Vision Transformer architectures without a `[cls]` token, we adapt the default VISSL evaluation recipe to utilize the average-pooled patch representation instead of the `[cls]` token. We therefore report the best linear evaluation number among the following representations: 1) the average-pooled patch representation of the last layer, 2) the concatenation of the last 4 layers of the average-pooled patch representations. We otherwise keep the linear-probing recipe identical.

ImageNet evaluations. To evaluate the I-JEPA on ImageNet [60], we adapt the VISSL recipe to use average pooled representations instead of the `[cls]` token. Following MAE [36], we use the LARS [72] optimizer with a batch-size of 16384, and train the linear probe for 50 epochs. We use a learning rate with a step-wise decay, dividing it by a factor of 10 every 15 epochs, and sweep three different reference learning rates $[0.01, 0.05, 0.001]$, and two weight decay values $[0.0005, 0.0]$.

A. 实现细节

A.1. 预训练

架构。I-JEPA 的上下文编码器、目标编码器和预测器均采用 Vision Transformer（ViT）[29]。两个编码器为标准 ViT，预测器则是轻量、较窄的 ViT。预测器嵌入维度固定为 384，自注意力头数与上下文编码器骨干相同。较小的 ViT-B/16 使用深度 6 的预测器；ViT-L/16、ViT-H/16、ViT-H/14 使用深度 12；ViT-G/16 使用深度 16。预训练不使用 [cls]令牌。评估时使用目标编码器，对输出平均池化生成全局图像表征。

优化。使用 AdamW [51]优化上下文编码器与预测器，默认批量 2048。预训练前 15 轮，学习率从 10^{-4} 线性升至 10^{-3}，随后按余弦日程降至 10^{-6}。遵循 [4, 18],，预训练期间权重衰减由 0.04 线性增至 0.4。目标编码器初始化权重与上下文编码器相同，此后通过指数移动平均更新 [4, 18, 23, 35, 37, 61]。遵循 [4, 18]，动量从 0.996 在预训练中线性增至 1.0。

遮蔽。默认抽取 4 个可能重叠的目标块，随机尺度范围（0.15, 0.2），纵横比范围（0.75, 1.5）；抽取 1 个上下文块，尺度（0.85, 1.0），纵横比为 1。随后从上下文掩码中移除与任意目标块重叠的区域。对小批量中每张图像独立抽取上下文和目标掩码。为高效批处理，同一 GPU 上全部上下文掩码大小相同，全部目标掩码大小也相同。掩码采样器通过在数据加载进程运行的批量整理函数，用几行 PyTorch [56] 代码高效实现。每次迭代，数据加载器返回一批图像，以及每张图像的上下文和目标掩码，标明两个视图需要保留的图块索引。

A.2. 下游任务

线性评估。评估 iBOT [79], 、DINO [18] 或 MAE [36], 等使用额外 [cls] 令牌的 Vision Transformer [29] 方法时，采用 VISSL [34] 默认配置，在 iNaturalist18 [65], 、CIFAR100 [45], 、Clevr/Count [42, 75], 、Clevr/Dist [42, 75], 和 Places205 [78]上评估全部模型。冻结编码器，报告以下表征中的最佳结果：1）最后一层的 [cls] 令牌；2）最后 4 层 [cls] 令牌的拼接。每种表征尝试两种头：1）线性头；2）先批归一化再线性头，报告最佳值。采用 VISSL [34]默认增强：随机缩放裁剪与水平翻转。但 Clevr/Count 和 Clevr/Dist 只用中心裁剪与水平翻转，因为随机裁剪会移除关键物体，干扰计数和距离估计。CIFAR100 图像缩放为 224×224 像素，以保持图块数量与预训练相同。

I-JEPA 的 ViT 不带 [cls] 令牌，因此调整 VISSL 默认方案，以平均池化图块表征代替 [cls] 令牌。报告以下表征中的最佳线性评估：1）最后一层平均池化图块表征；2）最后 4 层平均池化表征的拼接。其他线性探测设置相同。

ImageNet 评估。评估 I-JEPA [60], 时，调整 VISSL 方案，用平均池化表征代替 [cls] 令牌。遵循 MAE [36], ，采用 LARS [72] ，批量 16384，训练线性探测 50 轮。学习率每 15 轮降为十分之一，搜索三种参考学习率 $[0.01, 0.05, 0.001]$和两种权重衰减 $[0.0005, 0.0]$。

Low-shot evaluation. To evaluate our model on the ImageNet-1% low-shot task, we adapt the fine-tuning protocol of MAE [36].We fine-tune our ViT-L/H models for 50 epochs on ImageNet-1% with the AdamW optimizer and a cosine learning rate scheduler. We use a batch size of 512, a learning rate layer decay of 0.75 and 0.1 label smoothing. We use the default randaugment data-augmentations as in MAE. In contrast to the fine-tuning done with MAE, we do not use mixup, cutmix, random erasing or drop path. For the I-JEPA, we use a learning rate /weight decay of $3e^{-5}/5e^{-2}$ for the ViT-L/16, $3e^{-5}/4e^{-1}$ for the ViT-H/14 and $3e^{-5}/4e^{-1}$ for the ViT-H/16_{448}. Similar fine-tuning strategy for low-shot learning has been explored by Semi-VIT in the context of semi-supervised learning [16].

B. Broader Related Work

Self-supervised learning of visual representations with joint-embedding architectures is an active line of research [3, 10, 12, 18, 23, 24, 35, 37, 54, 69, 79]. These approaches train a pair of encoders to output similar embeddings for two or more views of the same image. To avoid pathological solution, many popular joint-embedding approaches use explicit regularization [5, 10, 18, 20] or architectural constraints [24, 35]. Collapse-prevention based on architectural constraints leverage specific network design choices to avoid collapse, for example, by stopping the gradient flow in one of the joint-embedding branches [20], using a momentum encoder in one of the joint-embedding branches [35], or using an asymmetric prediction head [8, 20, 35]. Recent work [62] attempts to theoretically understand (in certain simplified settings) how joint-embedding methods with architectural constraints avoid representation collapse without explicit regularization.

Typical regularization-based approaches to collapse prevention in joint-embedding architectures try to maximize the volume of space occupied by the representations. This is often motivated through the InfoMax [52] principle. Indeed, a long-standing conviction in unsupervised representation learning is that the resulting representations should be both maximally informative about the inputs, while also satisfying certain simplicity constraints [33,50]. The former objective is often referred to as the information-maximization principle (InfoMax), while the latter is sometimes referred to as the parsimony principle [52]. Such approaches to representation learning have been proposed for decades (e.g., [14]), where, historically, simplicity constraints were enforced by encouraging the learned representations to be sparse, low-dimensional, or disentangled, i.e., the individual dimensions of the representation vector should be statistically independent [33]. Modern approaches enforce the simplicity constraints coupled with InfoMax regularization through self-supervised loss terms [6, 40, 41, 44, 55, 64]. One example is the widespread view-invariance penalty [53], often coupled with with independence [10,74] or low-dimensionality constraints, e.g., by projecting representations on the unit hypersphere [20, 35, 37]. However, despite its proliferation, there have also been many criticisms of the InfoMax principle, especially since it is does not discriminate between different types of information (e.g, noise and semantics) [2]. Indeed, the sets of features we wish the model to capture are not always those with the highest marginal entropy (maximal information content).

Orthogonal to the contributions of invariance-based pretraining, another line of work attempts to learn representations by artificially masking parts of the input and training a network to reconstruct the hidden content [67]. Autoregressive models, and denoising autoencoders in particular, predict clean visual inputs from noisy views [8, 9, 19, 36, 67]. Typically, the goal is to predict missing inputs at a pixel level [29, 36, 70], or at a patch token-level, using a tokenizer [9, 68]. While these works demonstrate impressive scalability, they usually learn features at a low-level of semantic abstraction compared to joint-embedding approaches [4].

More recently, a set of approaches attempt to combine both joint-embedding architectures and reconstruction based approaches [30], wherein they combine an invariance pretraining loss with a patch-level reconstruction loss, as in the iBOT method [79]. Since view-invariance based approaches are typically biased towards learning global image representations, thereby limiting their applicability to other computer vision tasks, the idea is that adding local loss terms can improve performance on other popular tasks in computer vision [11, 26, 32]. The framework of contrastive predictive coding [55] is also closely related to this line of work on local loss terms. In the context of images [39], here the idea is to use a contrastive objective combined with a convolutional network to discriminate between overlapping image patch representations. Specifically, the goal is to encourage the representations of an image patch to be predictable of the image patches directly below it, while pushing away the representations of other patch views. In contrast to that work, the proposed I-JEPA method is non-contrastive and does not seek to discriminate between image patches. Rather, the goal is to predict the representations of various target blocks from a single context block. This is achieved with a Joint-Embedding Predictive Architecture, using a predictor network that is conditioned on positional embeddings corresponding to the location of the target block in the image. Qualitative experiments in Section 8 show that the predictor network in our architecture learns to correctly perform this local-to-local region feature mapping, and learns to correctly capture positional uncertainty in the image.

少样本评估。在 ImageNet-1% 上，改用 MAE [36]微调协议：ViT-L/H 在 1% 数据上用 AdamW 和余弦学习率调度微调 50 轮，批量 512，逐层学习率衰减 0.75，标签平滑 0.1。使用与 MAE 相同的默认 RandAugment，但不使用 mixup 、 cutmix 、随机擦除或随机深度。I-JEPA 的学习率／权重衰减为：ViT-L/16 用 $3e^{-5}/5e^{-2}$，ViT-H/14 用 $3e^{-5}/4e^{-1}$，ViT-H/16_{448}用 $3e^{-5}/4e^{-1}$。Semi-VIT 在半监督学习中也探索过类似少样本微调策略 [16]。

B. 更广泛的相关工作

使用联合嵌入架构进行视觉自监督表征学习，是活跃研究方向 [3,10, 12, 18, 23, 24, 35, 37, 54, 69, 79]。这些方法训练一对编码器，为同一图像两个或多个视图输出相似嵌入。为避免病态解，许多方法使用显式正则化 [5, 10, 18, 20] 或架构约束 [24, 35]。后者通过具体网络设计防坍塌，例如停止某个联合嵌入分支的梯度 [20], 、在一个分支使用动量编码器 [35], ，或使用非对称预测头 [8, 20, 35]。近期工作 [62] 尝试在某些简化条件下，理论解释带架构约束的联合嵌入方法如何无需显式正则化而避免表征坍塌。

联合嵌入架构中，典型的正则化防坍塌方法试图最大化表征所占空间的体积，通常以 InfoMax [52] 原则为依据。无监督表征学习长期认为，表征应尽可能包含输入信息，同时满足某些简单性约束 [33,50]。前者通常称信息最大化（InfoMax）原则，后者有时称简约原则 [52]。此类方法已提出数十年（如 [14]）；早期通过鼓励表征稀疏、低维或解耦来实现简单性，即表征向量各维统计独立 [33]。现代方法通过自监督损失项 [6, 40, 41, 44, 55, 64]将简单性约束与 InfoMax 正则化结合。例如，广泛使用的视图不变性惩罚 [53], 通常结合独立性 [10,74] 或低维约束，如将表征投影到单位超球面 [20, 35, 37]。但 InfoMax 虽然流行，也受到许多批评，尤其因为它不区分信息类型（如噪声与语义） [2]。我们希望模型捕获的特征，并不总是边缘熵最高、信息量最大的特征。

与不变性预训练独立的另一研究方向，是人为遮蔽部分输入、训练网络重建隐藏内容 [67]。自回归模型，尤其去噪自编码器，从带噪视图预测干净视觉输入 [8, 9, 19, 36, 67]，通常在像素级 [29, 36, 70], ，或借助分词器在图块令牌级 [9, 68]预测缺失输入。虽有良好扩展性，相比联合嵌入方法 [4]，它们通常学到语义抽象程度较低的特征。

最近一些方法尝试结合联合嵌入架构与重建方法 [30], ，将不变性预训练损失和图块级重建损失组合，如 iBOT [79]。视图不变性方法通常偏向全局图像表征，限制其他视觉任务的适用性，因此加入局部损失可能改善其他常见任务 [11, 26, 32]。对比预测编码框架 [55] 也与局部损失研究紧密相关。在图像中 [39], ，它以对比目标结合卷积网络，区分重叠图块表征：鼓励一个图块的表征能够预测正下方图块，同时排斥其他图块视图的表征。与之不同，I-JEPA 是非对比方法，不试图区分图块，而是从一个上下文块预测多个目标块表征。通过联合嵌入预测架构实现，预测器以目标块在图像中的位置嵌入为条件。第 8 节定性实验显示，预测器学会正确执行局部区域之间的特征映射，并正确捕获图像中的位置不确定性。

Targets		Context	
Scale	Freq.	Scale	**Top-1**
(0.075, 0.2)	4	(0.85, 1.0)	19.2
(0.1, 0.2)	4	(0.85, 1.0)	39.2
(0.125, 0.2)	4	(0.85, 1.0)	42.4
(0.15, 0.2)	4	(0.85, 1.0)	**54.2**
(0.2, 0.25)	4	(0.85, 1.0)	38.9
(0.2, 0.3)	4	(0.85, 1.0)	33.6

Table 8. **Ablation of the target block size for multi-block masking**. Linear evaluation on 1% ImageNet-1K (using only 1% of the available labels); ablating the multi-block target size during I-JEPA pretraining of a ViT-B/16 for 300 epochs. Predicting larger (semantic) blocks improves the low-shot accuracy as long as the context is sufficiently informative.

Targets		Context	
Scale	Freq.	Scale	**Top-1**
(0.15, 0.2)	4	(0.40, 1.0)	31.2
(0.15, 0.2)	4	(0.65, 1.0)	47.1
(0.15, 0.2)	4	(0.75, 1.0)	49.3
(0.15, 0.2)	4	(0.85, 1.0)	**54.2**

Table 9. **Ablation of the context size for multi-block masking**. Linear evaluation on 1% ImageNet-1K (using only 1% of the available labels); ablating the multi-block target size during I-JEPA pretraining of a ViT-B/16 for 300 epochs. Reducing the multi-block context size degrades the low-shot performance.

Targets		Context	
Scale	Freq.	Scale	**Top-1**
(0.15, 0.2)	1	(0.85, 1.0)	9.0
(0.15, 0.2)	2	(0.85, 1.0)	22.0
(0.15, 0.2)	3	(0.85, 1.0)	48.5
(0.15, 0.2)	4	(0.85, 1.0)	**54.2**

Table 10. **Ablation of the targets number for multi-block masking**. Linear evaluation on 1% ImageNet-1K (using only 1% of the available labels); ablating the multi-block number of targets during I-JEPA pretraining of a ViT-B/16 for 300 epochs. Increasing the number of target blocks improve the low-shot accuracy.

C. Additional Ablations

This section follows the same experimental protocol as Section 9. We report the result of a linear probe with a frozen backbone, trained on the low-shot 1% ImageNet-1K benchmark.

Multiblock masking strategy. We present an extended ablation of the multiblock masking strategy where we change the targets block scale (Table 8), the context scale (Table 9) and the number of target blocks (Table 10). We train a ViT-B/16 for 300 epochs using I-JEPA with various `multi-block` settings and compare performance on the 1% ImageNet-1K benchmark using a linear probe. In short, we find that it is important to predict several relatively large (semantic) target blocks, and to use a sufficiently informative (spatially distributed) context block.

Masking at the output of the target-encoder. An important important design choice in I-JEPA is that the target blocks are obtained by masking the *output* of the target-encoder, not the input. Table 11 shows the effect of this design choice on the semantic level of the learned representations when pretraining a ViT-H/16 using I-JEPA for 300 epochs. In the case where masking is applied to the input, we forward-propagate through the target-encoder once for each target region. Masking the output of the target-encoder during pretraining results in more semantic prediction targets and improves linear probing performance.

目标块		上下文	
尺度	频率	尺度	**Top-1**
(0.075, 0.2)	4	(0.85, 1.0)	19.2
(0.1, 0.2)	4	(0.85, 1.0)	39.2
(0.125, 0.2)	4	(0.85, 1.0)	42.4
(0.15, 0.2)	4	(0.85, 1.0)	**54.2**
(0.2, 0.25)	4	(0.85, 1.0)	38.9
(0.2, 0.3)	4	(0.85, 1.0)	33.6

表 8. 多块掩码目标大小消融。ViT-B/16 用 I-JEPA 预训练 300 轮，仅用 1% ImageNet-1K 标签线性评估。只要上下文足够有信息，预测更大、具有语义的块可提高少样本准确率。

目标块		上下文	
尺度	频率	尺度	**Top-1**
(0.15, 0.2)	4	(0.40, 1.0)	31.2
(0.15, 0.2)	4	(0.65, 1.0)	47.1
(0.15, 0.2)	4	(0.75, 1.0)	49.3
(0.15, 0.2)	4	(0.85, 1.0)	**54.2**

表 9. 多块掩码上下文大小消融。ViT-B/16 用 I-JEPA 预训练 300 轮，仅用 1% ImageNet-1K 标签线性评估。缩小多块上下文降低少样本性能。

目标块		上下文	
尺度	频率	尺度	**Top-1**
(0.15, 0.2)	1	(0.85, 1.0)	9.0
(0.15, 0.2)	2	(0.85, 1.0)	22.0
(0.15, 0.2)	3	(0.85, 1.0)	48.5
(0.15, 0.2)	4	(0.85, 1.0)	**54.2**

表 10. 多块掩码目标数量消融。ViT-B/16 用 I-JEPA 预训练 300 轮，仅用 1% ImageNet-1K 标签线性评估。增加目标块数提高少样本准确率。

C. 额外消融

本节遵循第 9 节实验协议：冻结骨干，在少样本 1% ImageNet-1K 基准上训练线性探测并报告结果。

多块掩码策略。扩展消融改变目标块尺度（表 8）、上下文尺度（表 9）和目标块数（表 10）。不同设置的 ViT-B/16 用 I-JEPA 预训练 300 轮，再在 1% ImageNet-1K 上线性探测。我们发现，预测多个较大、具有语义的目标块，以及使用足够有信息、空间广泛分布的上下文块十分重要。

在目标编码器输出处遮蔽。I-JEPA 的重要设计是遮蔽目标编码器输出，而非输入，以取得目标块。表 11 展示 ViT-H/16 预训练 300 轮时该选择对语义层次的影响。若遮蔽输入，每个目标区域都需单独通过目标编码器。预训练时遮蔽输出，可产生更语义化的预测目标并提升线性探测性能。

Target Masking	Arch.	Epochs	Top-1
Output	ViT-H/16	300	**67.3**
Input	ViT-H/16	300	56.1

Table 11. **Ablating masking output of target encoder**. Linear evaluation on ImageNet-1K using only 1% of the available labels; ablating the effect of masking the target-encoder output during I-JEPA pretraining of a ViT-H/16 for 300 epochs. Masking the output of the target-encoder during pretraining significantly improves the linear probing performance of the pretrained representations.

Predictor depth. We examine the impact of the predictor depth on the downstream low-shot performance in Table 12. We pretrain a ViT-L/16 for 500 epochs using either a 6-layer predictor network or a 12-layer predictor network. The model pretrained using a deeper predictor shows a significant improvement in downstream low-shot performance compared to the model pretrained with a shallower predictor.

Predictor Depth	Arch.	Epochs	Top-1
6	ViT-L/16	500	64.0
12	ViT-L/16	500	**66.9**

Table 12. **Ablating the predictor depth**. Linear evaluation on ImageNet-1K using only 1% of the available labels; ablating the effect of masking the predictor depth for a ViT-L/16 pretrained for 500 epochs. Increasing the predictor depth leads to significant improvement of the linear probe performance of the pretrained representations.

Weight decay. In Table 13, we evaluate the impact of weight-decay during pretraining. We explore two weight decay strategies: linearly increase the weight-decay from 0.04 to 0.4 or use a fix weight-decay of 0.05. Using a smaller weight decay during pretraining improves the downstream performance on ImageNet-1% when fine-tuning. However, this also leads to a degradation of performance in linear evaluation. In the main paper, we use the first weight decay strategy as it improves the performances in linear evaluation downstream tasks.

Weight Decay	Arch.	Epochs	ImageNet-1%	ImageNet Linear-Eval
0.04 → 0.4	ViT-L/16	600	69.4	**77.8**
0.05	ViT-L/16	600	**70.7**	76.4

Table 13. **Ablating the pretraining weight-decay**. We compare our default pretraining weight decay strategy where we linearly increase the weight-decay from 0.04 to 0.4 to using a fix weight decay of 0.05. Using a smaller weight-decay during pretraining can improve the fine-tuning performance on ImageNet-1%, However, it also leads to a drop of performance in linear evaluation.

Predictor width. We explore the impact of the predictor width in Table 14. We compare I-JEPA using a ViT-L encoder and a predictor with 386 channels to a similar model using a predictor with 1024 channels. Note that the ViT-L encoder has 1024 channels. Using a bottleneck in the predictor width improves the downstream performance on ImageNet 1%.

Predictor Width	Arch.	Epochs	Top-1
384	ViT-L/16	600	**70.7**
1024	ViT-L/16	600	68.4

Table 14. **Ablating the predictor width.** We reports results on ImageNet-1K 1% using fine-tuning. We compare two predictors having a width of either 384 or 1024. Note the I-JEPA encoder is a ViT-L with 1024 channels. Having a width bottleneck in the predictor improves the downstream performances.

D. Finetuning on the full ImageNet

In this section, we report performance on I-JEPA when fine-tuning on the full ImageNet dataset. We focus on the ViT-H/16_{448} as this architecture achieves state-of-art performance with MAE [36].

目标遮蔽	架构	轮数	Top-1
输出	ViT-H/16	300	**67.3**
输入	ViT-H/16	300	56.1

表 11. 目标编码器输出遮蔽消融。ViT-H/16 预训练 300 轮，仅用 1% ImageNet-1K 标签线性评估。预训练时遮蔽目标编码器输出显著提高表征的线性探测性能。

预测器深度。表 12 研究下游少样本性能。ViT-L/16 分别使用 6 层或 12 层预测器预训练 500 轮；较深预测器显著提高下游性能。

预测器深度	架构	轮数	Top-1
6	ViT-L/16	500	64.0
12	ViT-L/16	500	**66.9**

表 12. 预测器深度消融。ViT-L/16 预训练 500 轮，仅用 1% ImageNet-1K 标签线性评估。增加预测器深度显著提高预训练表征的线性探测性能。

权重衰减。表 13 比较预训练策略：从 0.04 线性增至 0.4，或固定 0.05。较小衰减提高 ImageNet-1% 微调性能，但降低线性评估性能。正文采用第一种策略，因为它改善下游线性评估。

权重衰减	架构	轮数	ImageNet-1%	ImageNet 线性评估
0.04 → 0.4	ViT-L/16	600	69.4	**77.8**
0.05	ViT-L/16	600	**70.7**	76.4

表 13. 预训练权重衰减消融：默认从 0.04 线性增至 0.4，与固定 0.05 比较。较小衰减可改善 ImageNet-1% 微调，但降低线性评估。

预测器宽度。表 14 比较 ViT-L 编码器搭配 386 通道预测器，与 1024 通道预测器。ViT-L 编码器有 1024 通道。预测器宽度瓶颈改善了 ImageNet 1% 下游性能。

预测器宽度	架构	轮数	Top-1
384	ViT-L/16	600	**70.7**
1024	ViT-L/16	600	68.4

表 14. 预测器宽度消融。ImageNet-1K 1% 微调比较 384 与 1024 宽度的预测器。编码器为 1024 通道 ViT-L。预测器宽度瓶颈提升下游性能。

D. 完整 ImageNet 上微调

本节报告完整 ImageNet 微调的 I-JEPA 性能，重点关注 ViT-H/16_{448}，因为该架构在 MAE [36]中达到最佳性能。

We use a fine-tuning protocol similar to MAE. Specifically, we fine-tune our model for 50 epochs using AdamW and a cosine learning rate schedule. The base learning rate is set to 10^{-4} and the batch size to 528. We train using mixup [76] set to 0.8, cutmix [73] set to 1.0, a drop path probability of 0.25 and a weight decay set to 0.04. We also use a layer decay of 0.75. Finally, we use the same rand-augment data-augmentations as MAE,

Table 15 reports the fine-tuning results. I-JEPA achieves 87.1 top-1 accuracy. Its performance is less than 1% away from the best MAE model despite I-JEPA being trained for 5.3 times less epochs than MAE. This result demonstrates that I-JEPA is competitive when fine-tuning on the full ImageNet dataset.

Method	Arch.	Epochs	Top-1
MAE [36]	ViT-H/14_{448}	1600	**87.8**
I-JEPA	ViT-H/16_{448}	300	87.1

Table 15. **Finetuning on the full ImageNet dataset**. I-JEPA achieves competitive performance. I-JEPA is close to MAE approach despite I-JEPA being trained for 5.3 times less epochs than MAE.

采用类似 MAE 的微调协议：AdamW 和余弦学习率日程，微调 50 轮，基础学习率 10^{-4}，批量 528。mixup [76]为 0.8，cutmix [73] 为 1.0，随机深度概率 0.25，权重衰减 0.04，逐层学习率衰减 0.75。采用与 MAE 相同的 RandAugment，

表 15 给出微调结果：I-JEPA top-1 准确率 87.1%。尽管预训练轮数比 MAE 少 5.3 倍，与最佳 MAE 的差距仍不到 1%。说明完整 ImageNet 微调具有竞争力。

方法	架构	轮数	Top-1
MAE [36]	ViT-H/14_{448}	1600	**87.8**
I-JEPA	ViT-H/16_{448}	300	87.1

表 15. 完整 ImageNet 数据集上的微调。I-JEPA 表现具有竞争力，接近 MAE，尽管预训练轮数比 MAE 少 5.3 倍。

E. RCDM Visualizations

To visualize the representations of a pretrained neural network in pixel space, we use the RCDM framework [13]. The RCDM framework trains a decoder network h_ω, comprising a generative diffusion model, to reconstruct an image $\boldsymbol{x}$ from the representation vector of that image $\boldsymbol{s}_x$ and a noisy version of that image $\hat{\boldsymbol{x}} := \boldsymbol{x} + \epsilon$, where ϵ is an additive noise vector. Concretely, the decoder objective is to minimize the loss function $\|h_\omega(\hat{\boldsymbol{x}}, \boldsymbol{s}_x) - \epsilon\|$. We train each RCDM network for 300,000 iterations using the default hyperparameters [13]. After training the decoder, one can subsequently feed the representation vector of an unseen test image $\boldsymbol{s}_y$ into the decoder along with various random noise vectors to generate several pixel-level visualizations of the representation, thus providing insight into the features captured in the representations of the pretrained network. Qualities that are common across samples represent information that is contained in the representation. On the other hand, qualities that vary across samples represent information that is not contained in the representations

In Figure 6, the visualizations are obtained by feeding the average-pooled output of the predictor, conditioned on a specific target region, into the decoder network, along with various random noise vectors. In Figures 7 and 8, the visualizations are obtained by feeding the average-pooled output of the target-encoder into the decoder network, along with various random noise vectors.

E. RCDM 可视化

为在像素空间可视化预训练网络表征，使用 RCDM 框架 [13]。它训练由生成式扩散模型组成的解码器网络 h_ω，根据图像表征向量 $\boldsymbol{s}_x$ 和带噪图像$\hat{\boldsymbol{x}}$ 重建图像 $\boldsymbol{x}$：$= \boldsymbol{x} + \epsilon$，其中 ϵ 是加性噪声向量。具体而言，解码器最小化损失 $\|h_\omega(\hat{\boldsymbol{x}}, \boldsymbol{s}_x) - \epsilon\|$。每个 RCDM 网络按默认超参数 [13]训练 300,000 次迭代。训练后，将未见测试图像的表征向量 $\boldsymbol{s}_y$ 和不同随机噪声向量送入解码器，生成多幅像素级可视化，了解预训练网络所捕获的特征。样本共有的属性表示表征中包含的信息，样本间变化的属性则表示表征未包含的信息。

E.1. Encoder Visualization

In Figure 7, we visualize the average-pooled I-JEPA representations at the output of our ViT-H/14 target-encoder. The first column contains the original image, while subsequent columns contain synthetic samples obtained by feeding the average-pooled representation of the image into the decoder along with various random noise vectors. Figure 7 suggests that the I-JEPA target-encoder is able to correctly capture the high-level information regarding objects and their poses, while discarding low-level image details and background information.

Figure 8 shows similar visualizations, but when using an MSN [4] pretrained ViT-L/7 target-encoder to compute the image representations. The MSN method trains a context- and target-encoder using a Joint-Embedding Architecture to enforce invariance of global image representations to various hand crafted data augmentations and missing patches. While the MSN pretrained network is able to capture high level semantic information about the image in the first column, it also exhibits higher variability in the generated samples, e.g., variability in the object pose, object scale, and number of instances. In short, the MSN pretrained discards much of the local structure in the image, which is in stark contrast to I-JEPA, which retains information about much of the local structure in the input image.

图 6 将以特定目标区域为条件的预测器平均池化输出，与不同随机噪声送入解码器。图 7、8 则将目标编码器平均池化输出与随机噪声送入解码器。

E.1. 编码器可视化

图 7 可视化 ViT-H/14 目标编码器输出的平均池化 I-JEPA 表征。首列原图，后续列将平均池化表征和不同随机噪声输入解码器得到合成样本。结果表明，目标编码器正确捕获物体及姿态的高层信息，同时舍弃低层细节和背景。

图 8 类似，但图像表征由 MSN [4] 预训练的 ViT-L/7 目标编码器计算。MSN 通过联合嵌入架构训练上下文与目标编码器，使全局表征对手工增强和缺失图块保持不变。它虽能捕获首列图像的高层语义，但生成样本变化更大，例如物体姿态、尺度和实例数量。MSN 舍弃了大量局部结构，与保留许多局部输入结构的 I-JEPA 形成鲜明差别。

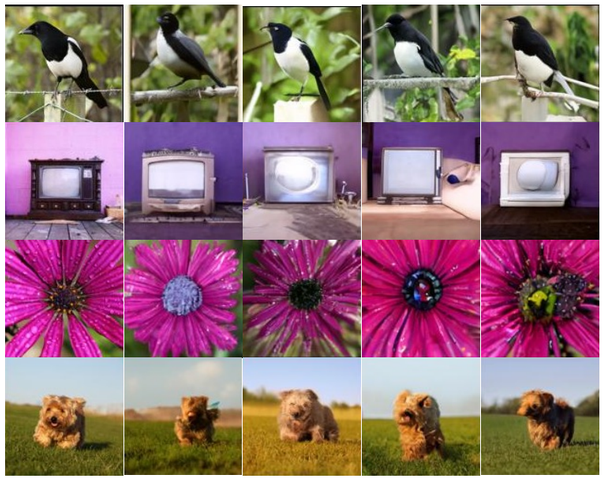


Figure 7. **Visualization of I-JEPA target-encoder representations**. For each image: first column contains the original image; subsequent columns contain samples from a generative model decoding the average-pooled output of a pretrained I-JEPA target-encoder. Qualities that are common across samples represent information that contained is in the I-JEPA representation. I-JEPA is able to correctly capture the high-level information regarding objects and their poses. Qualities that vary across samples represent information that is not contained in the representation. I-JEPA encoder discards the precise low-level details as well as background information.



Figure 8. **Visualization of MSN target-encoder representations**. For each image: first column contains the original image; subsequent columns contain samples from a generative model decoding the output of a frozen MSN encoder [4]. Qualities that are common across samples represent information that is contained in the representation. Qualities that vary across samples represent information that is not captured by MSN. Compared to I-JEPA, MSN samples show higher variability. MSN retains less information from the input. In particular, it discards global structure information such as the object pose or even number of instances.

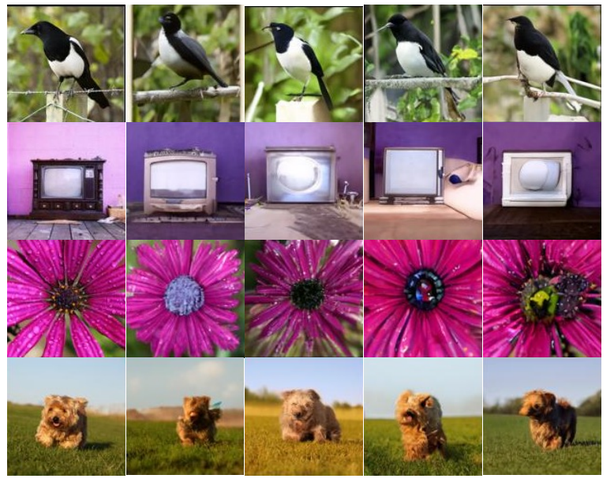


图 7. I-JEPA 目标编码器表征可视化。首列为原图，后续列为生成模型解码预训练目标编码器平均池化输出的样本。样本共有属性代表表征所含信息。I-JEPA 正确捕获物体及其姿态等高层信息；变化属性代表未含信息。编码器舍弃精确低层细节和背景。



图 8. MSN 目标编码器表征可视化。首列原图，后续列为生成模型解码冻结 MSN 编码器 [4]输出的样本。共有属性是表征所含信息，变化属性是 MSN 未捕获信息。相比 I-JEPA，MSN 样本变化更大、保留输入信息更少，尤其舍弃物体姿态甚至实例数量等全局结构。